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PERFORMANCE SPECIFICATION
RADIO SET AN/GRC-103(V)3

This performance specification is approved for use by the USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This performance specification covers the requirements for the design and construction of one type of equipment, designated Radio Set AN/GRC-103(V)3 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2). Handbooks are for guidance only and therefore are not mandatory.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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HANDBOOKS

DEPARTMENT OF DEFENSE

MIL-HDBK-454	- Standard General Requirements for Electronic Equipment
MIL-HDBK-781	- Reliability Testing for Engineering Development, Qualification and Production

MIL-STD-810 Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of the above specifications, standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4d, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

USA CECOM

SC-A-46439	- List of Accessories for Package Tester
GL-SM-B-323451	- Mechanical. Gages for Radio Set AN/GRC-103(V)3
SC-DL-424153	- Adapter, Connector UG-1375( )/U
DL-SM-B-696006	- Receiver, Radio R-1329( ) (P)/GRC-103(V)
SM-A-696070	- Test Procedure - Mast AB-952( )/GRC-103(V)
DL-SM-B-696081	- Antenna AS-1854 ( )/GRC-103(V)
DL-SM-B-696083	- Amplifier-Converter AM-4318( )/GRC-103(V)
DL-SM-B-696111	- Amplifier-Frequency Multiplier AM-4322( )/GRC-103(V)
DL-SM-B-696151	- Mast AB-952( )/GRC-103(V)

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DL-SM-B-696171	- Transmitter, Radio T-983( )(P)/GRC-103(V)
DL-SM-B-696175	- Rcvr - Trans, Order Wire, RT-773( )/GRC-103(V)
DL-SM-B-696203	- Cable Assy., R.F. CG-3443( )/U
DL-SM-B-696207	- Cable Assy., R.F. CG-3444( )/U
DL-SM-B-696210	- Cable Assy., S.P. Elec., CX-10763( )/GRC-103(V)
SM-A-696387	- Alignment Procedure - Duplexer Subassembly
SM-A-696394	- Test Procedure - Control Indicator
SM-A-696396	- Alignment Procedure - Frequency Multiplier
SM-A-696397	- Alignment Procedure - Band Pass Filter
SM-A-696443	- Test Specification - Amplifier - Frequency Multiplier
SM-A-696455	- Antenna AS-1854( )/GRC-103(V)
SM-A-696763	- Mixer Stage, Frequency - Alignment Procedure
SM-E-697509	- Housing, Amplifier Frequency Multiplexer
SM-A-698002	- Test Specification - Intermediate Frequency Amplifier
SM-A-698031	- Test Specification - Electrical Frequency Limiter-Discriminator
SM-A-698035	- Test Specification - Video Amplifier
SM-D-698201	- Cable Assy, Radio Frequency
SM-D-698202	- Cable Assy, Radio Frequency
SM-D-698204	- Cable Assy, Radio Frequency
SM-A-698241	- Test Specification - Pulse Form Restorer
SM-A-698317	- Test Specification - Panel Monitor

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DRAWINGS (Continued)

USA CECOM

SM-A-698363	- Test Specification - Power Supply
SM-A-698372	- Test Specification - Power Supply
SM-A-698610	- Test Specification - Amplifier - Frequency Multiplier
SM-A-865041	- Test Specification - Electrical Frequency Synthesizer
SM-A-698643	- Alignment Procedure - Signal Level Control - Monitor
SM-A-698652	- Alignment Procedure - Frequency Multiplier Group
SM-A-698656	- Alignment Procedure - Amplifier - Converter, AM-4318( )/GRC-103(V)
SM-A-698657	- Alignment Procedure - Amplifier - Frequency Multiplier
SM-A-698658	- Test Procedure - Electrical Dummy Load
SM-A-698659	- Alignment Procedure - Radio Frequency Amplifier
SM-A-698660	- Test Procedure - AM-4322( )/GRC-103(V)
SM-A-698664	- Test Specification - Amplifier Monitor
SM-A-698685	- Test Specification - Distribution Box
SM-A-698695	- Test Specification - Monitor Panel
SM-A-698759	- Test Specification - Distribution Panel
SM-A-698776	- Test Specification - Alarm Control

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SM-A-698781 - Alignment Procedure - Frequency Multiplier

DRAWINGS (Continued)

USA CECOM

SM-A-698789 - Test Specification - Distribution Box

SM-A-698790 - Test Specification - Monitor Panel

SM-A-698805 - Test Specification - Telephone Signal Converter

SM-A-698806 - Test Specification - Amplifier Assembly

SM-A-698807 - Test Specification - RT-773( )/GRC-103(V)

SM-A-883588 - Test Specification - Radio Receiver R-1329(P)( )/GRC-103(V)

SM-A-883590 - Test Specification - Transmitter T-983(P)( )/GRC-103(V)

SM-A-764245 - Test Specification - Radio Set AN/GRC103( )(V)3

SM-A-764246 - Test Specification - Radio Set AN/GRC103( )(V)3 (Transmitter Section)

SM-A-764247 - Test Specification - Radio Set AN/GRC103( )(V)3 (Receiver Section)

SM-A-698826 Test Specification- Radio Receiver R-1329()(P)/GRC-103(V)

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

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ANSI C63.14 - Electromagnetic Capability (EMC), etc; Dictionary for Technologies of

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018).

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish sample units for first article inspection in accordance with 4.2.

3.2 Materials. The contractor shall select materials that are fully capable of meeting all of the operational and environmental requirements specified herein. The materials specified in the applicable drawings are recommended, but are not mandatory. Selection of the class, grade or type part shall be that the material will be able to perform its intended function when it is assembled. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard. Recovered materials shall be used to the maximum extent possible.

3.3 Description. Radio Set AN/GRC-103(V)3 provides for multichannel, line-of-sight, two-way communications in the frequency range of 695-1000 MHz (Channels 990-1600). This version is intended primarily for use as a radio link in a communications network which may include carrier telephone and teletypewriter sets. The radio set is used in pairs: one at each end of the line-of-sight path. The radio set may be used with 6/12 or 24 channel Pulse Code Modulation (PCM) equipment or with 4 channel frequency division modulation (FDM) when associated with approximate interface equipment. As currently designed, the set provides channel assignments on each half and full megacycle points (1/2 MHz spacing).

3.4 Design. The radio set and associated subassemblies and components shall meet the dimensions and dimensional tolerances of the drawings list in the following Data List drawings:

DL-SM-B- 696006 Receiver, Radio R-1329()(P)/GRC-103(V)

DL-SM-B-696081 - Antenna AS-1854 ( )/GRC-103(V)

DL-SM-B-696083 - Amplifier-Converter AM-4318( )/GRC-103(V)

DL-SM-B-696111 - Amplifier-Frequency Multiplier AM-4322( )/GRC-103(V)

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DL-SM-B-696151	- Mast AB-952( )/GRC-103(V)
DL-SM-B-696171	- Transmitter, Radio T-983( )(P)/GRC-103(V)
DL-SM-B-696175	- Rcvr - Trans, Order Wire, RT-773( )/GRC-103(V)
DL-SM-B-696203	- Cable Assy., R.F. CG-3443( )/U
DL-SM-B-696207	- Cable Assy., R.F. CG-3444( )/U
DL-SM-B-696210	- Cable Assy., S.P. Elec., CX-10763( )/GRC-103(V)
SC-DL-424153	- Adapter, Connector UG-1375()/U

3.5 Performance characteristics. The performance characteristics for the Radio Set AN/GRC-103(V)3 will be defined within the individual system requirements for the following set components. These components are the Transmitter section (3.5.1), Receiver section (3.5.2), Terminal section (3.5.3), Receiver Transmitter Order Wire (3.5.4), Antenna (3.5.5), and the Mast (3.5.6).

3.5.1 Transmitter section.

3.5.1.1 Output power. The normal transmitter RF output power, without the RF duplexer, shall be not less than 20 watts at any frequency in the range of 695 to 1000 MHz.

3.5.1.2 Low power alarm. A visual and audible alarm shall be provided in the transmitter. The visual alarm shall indicate any change in RF output power that is greater than 8 dB + 2 dB - 0 dB below the nominal power output rating of the transmitter. The switch associated with the audible alarm is included such that; in one switch position the alarm shall sound when the visual alarms are lighted as in the case of a failure; in the second switch position, the alarm sounds when the light is not lighted. The audible alarm thus indicates a change in state of the visual signal.

3.5.1.3 Output tube dissipation. The transmitter output power shall not decrease below specification limits after:

a. Running the transmitter for 15 minutes with its output terminals short-circuited.

b. Running the transmitter for 15 minutes with its output terminals open-circuited.

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c. Running the transmitter for 30 minutes without drive.

3.5.1.4 Frequency setability. It shall be possible to set the transmitter to any desired frequency at 0.5 MHz intervals in the frequency range of 695 to 1000 MHz, with an accuracy of better than 0.002% corresponding to 20 Hz per MHz.

3.5.1.5 Transmitter frequency response. The overall frequency response of the transmitter when a modulation signal to produce 100 kHz peak deviation is applied to the video input shall conform to Table I.

TABLE I. Frequency response (transmitter).

Relative attenuation (dB) ref 10 kHz			
Frequency	Optimum	Minimum	Maximum
000.125	01.0	0.5	01.5
000.200	00.5	0	01.0
000.250	00.2	0	00.4
001.000	0	-0.2	00.2
010.00	0 (Ref)	---	---
030.00	0	-0.2	00.2
060.00	0.1	0	00.3
120.00	0.15	0	00.5
240.00	0.7	0.5	01.3
360.00	1.5	0.6	02.3
480.00	2.7	1.9	04.2
600.00	4.2	3.4	06.2
720.00	6.0	5.3	08.8
840.00	8.2	7.5	11.5
960.00	10.7	8.5	15.0

3.5.1.6 Spurious outputs. When measured with duplexer, the power output of any harmonic shall be at least 80 dB below that of the fundamental and the power output of any non-harmonic extraneous emission of a band outside 15 MHz of the carrier shall be at least 80 dB below that of the fundamental. All spurious outputs over the frequency range of ± 15 MHz shall be suppressed at least 60 dB, except for the range occupied by the modulation, where the spurious outputs shall not exceed the level of the modulation envelope.

3.5.1.7 FDM modulation requirements. With a 1 kHz sinusoidal signal of 191 mV applied to the transmitter video input, and the input control adjusted to give a reading of 50% on the panel meter, the transmitter peak deviation shall be between 65 kHz and 97 kHz. This requirement shall be met with the meter switch in the FDM position.

3.5.2 Receiver section.

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3.5.2.1 Frequency response shaping. The overall frequency response of the receiver shall conform to Table II.

TABLE II Frequency response (receiver).

Relative attenuation (dB) ref 10 kHz			
Frequency	Optimum	Minimum	Maximum
000.125	01.0	-0.2	02.5
000.200	00.5	-0.2	01.0
000.250	00.2	-0.2	00.4
001.000	0	-0.1	00.1
010.00	0 (Ref)	---	---
030.00	0	-0.1	00.1
060.00	0.15	0.05	00.25
120.00	0.6	0.2	00.9
180.00	1.3	0.8	01.8
240.00	2.3	1.8	02.8
360.00	5.1	4.4	05.8
480.00	9.0	8.2	10.0
600.00	14.0	12.5	15.5
720.00	20.2	18.4	22.6
840.00	27.5	25.3	31.0
960.00	36.0	33.0	40.0

(NOTE: Frequency response shall be measured when a frequency modulated signal of constant deviation (100 kHz peak) is applied to the receiver antenna terminal as indicated at the video output terminal.

3.5.2.2 Noise figure. The noise figure of the receiver, including the losses due to the receiver protection circuit and the duplexer, shall not exceed 10 dB between 695 and 1000 MHz.

3.5.2.3 Demodulation. For received signals deviated ± 300 kHz, ± 75 kHz and ± 45 kHz, the video outputs shall be 0.265 V ± 0.03 V RMS at the VIDEO connector, 0.189 ± 0.02 V RMS at the FDM connector and -10 dBm ± 2 dB (600 ohms) at the OW Connector, respectively.

3.5.2.4 Received signal monitoring. A panel meter deflection of 60% or more shall be obtained at a received signal level of -50 dBm and 10% or more at a level of -95 dBm.

3.5.2.5 IF bandwidth. The IF selectivity shall have a nominal 3 dB bandwidth of 750 kHz with the response down at least 60 dB at ± 2.1 MHz or more than resonance.

3.5.2.6 Receiver protection. The receiver shall be protected from damage which could result from its own transmitter throughout all receiver/transmitter frequency ranges. The radio set circuitry shall thus provide protection against external signals at any frequency having power levels to +47 dBm under all conditions of receiver tuning. Protection shall be provided against

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pulse modulation signals of any frequency of 1 kW peak power, 1 μs duration, 0.001 duty cycle, with a rise time of 10 ns from 10-90% amplitude.

3.5.2.7 Spurious response. The ratio of desired signal response to any spurious response shall be as follows:

a. For a maximum of 6 discrete responses, the ratio shall be at least 80 dB.

b. For all other responses, the ratio shall be at least 100 dB.

3.5.3 Terminal section (system requirements).

3.5.3.1 Input impedance. The impedance at the video input to the transmitter shall be 91 ohms ± 10% resistive to at least 480 kHz and 91 ohms ± 15% to at least ± 800 kHz.

3.5.3.2 Output impedance. The output impedance of the receiver shall be 51 ohms + 10% resistive for the 24 channel PCM output to at least 800 kHz and for the regenerated 6/12 channel PCM output such that when terminated in a resistive load of 91 ohms ± 5%, the waveform is as specified in 3.5.3.7.1.

NOTE: All received RF signal levels are measured at the antenna terminals of the duplexer.

3.5.3.3 Frequency response (PCM). The overall frequency response between transmitter video input and receiver video output relative to the response at 10 kHz shall conform to Table III.

TABLE III. Frequency response (PCM).

Relative attenuation (dB) ref 10 kHz			
Frequency	Optimum	Minimum	Maximum

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000.125	02.0	1.0	03.0
000.200	01.0	0	02.0
000.250	00.4	0	00.8
000.500	0	-0.1	+0.1
001.000	0	-0.1	+0.1
010.00	0 (Ref)	---	---
030.00	0	-0.2	00.2
060.00	0.2	0.1	00.4
120.00	0.7	0.5	01.0
180.00	1.6	1.3	02.2
240.00	2.9	2.5	04.0
360.00	6.5	5.7	08.0

TABLE III. Frequency response (PCM) - Continued.

Relative attenuation (dB) ref 10 kHz			
Frequency	Optimum	Minimum	Maximum
480.00	11.7	10.2	13.5
600.00	18.2	16.0	21.0
720.00	26.2	23.1	30.5
840.00	35.7	31.7	41.0
960.00	46.7	41.2	52.5

3.5.3.4 Pulse Code Modulation (PCM) requirements. An input of one volt (peak-to-peak) shall deviate the transmitter ± 300 kHz peak when the modulation sensitivity is adjusted to its maximum. The metering circuitry shall enable the operator to adjust the peak deviation of the transmitter to ± 180 kHz and ± 300 kHz with an accuracy of +2.2 and -1.25 dB with inputs of 1 to 4 volts (peak to peak). The transmitter shall be considered to have met this requirement if, when tested in accordance with the 12 and 24 CH PCM metering sections of the test specified in 4.19.3, it meets the specifications thereof.

3.5.3.5 Signal to noise (PCM). The transmitter shall be tuned to any of its operating frequencies and operate at rated RF power output. A 10 kHz sinusoidal signal shall be applied at the video input terminals of the transmitter and the transmitter shall be adjusted for a peak deviation of plus and minus 180 kHz. The receiver output, when the receiver input is coupled to the transmitter through sufficient attenuation, shall be 0.45V ± 0.15 peak to peak at the unregenerated output terminals. Under these conditions, the ratio of average detected signal power to the average noise power at the unregenerated video output terminals of the receiver as measured with a flat weighted meter shall be at least 12 dB when the receiver RF signal is -94 dBm and at least 31 dB when the received RF signal is -74 dBm. With the transmitter adjusted for rated peak deviated (plus and minus 3000 kHz), the receiver unregenerated video output shall be 3/4 ± 1/4 V peak-to-peak and the signal-to-noise ratio shall be at least 16 dB when the received RF signal is -94 dBm and at least 35 dB when the received signal is -74 dBm.

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3.5.3.6 Distortion (PCM). The radio set shall be arranged as in 3.5.3.5 with the receiver coupled to the transmitter through sufficient attenuation so that the RF input at the receiver input terminals (including duplexer) is -54 dBm. A sinusoidal signal at any frequency within the range of 250 Hz to 240 kHz shall be applied at the video input terminals of the transmitter at a level of 2 V peak-to-peak. The transmitter shall be adjusted for rated deviation and the receiver output shall be 3/4 plus or minus 1/4 V peak-to-peak at the unregenerated video output terminals. Under these conditions the total harmonic distortion measured at the unregenerated video output terminals shall not exceed a level 26 dB below the fundamental.

3.5.3.7 Pulse regeneration (PCM). The receiver output circuits shall generate new pulses whose amplitude, timing and pulse shape are derived from the 6 channel or 12 channel signals briefly described in Table IV.

TABLE IV. Pulse regeneration.

PCM characteristics		
Number of channels	6	12
Signal Bit rate	(Full-baud, unipolar binary) 288	(Full-baud, unipolar binary) 576
Pulse width at half amplitude	3.472	1.736

3.5.3.7.1 Regenerated PCM. The regeneration shall eliminate all noise peaks less than 40% of the pulse amplitude for the range of unregenerated video output levels corresponding to input deviations of +145 kHz peak minimum to ± 230 kHz peak maximum. The regenerated signal amplitude shall be -0.1 V ± 0.1 V for the “1” condition and -2.0 V ± 0.20 V for the “0” condition, delivered into a load of 91 ohms ± 5%, with no reversal of phase from the transmitter input or unregenerated video output. Pulse width shall be full interval with 10% to 90% rise time 10 ns minimum, 50 ns maximum. The 50% point shall be 0 ns minimum, 80 ns maximum after the 50% point of the rising edge of the timing signal specified in 3.4.3.7.2.

3.5.3.7.2 Timing signal. Timing signals shall be extracted from the incoming pulse train and shall accompany the regenerated PCM to the next unit. The timing signal amplitude shall be 2V ± 0.2 V DC isolated, delivered into a load of 91 ohms ± 5%. Pulse width at the 50% amplitude point shall be 80 ns minimum, 150 ns maximum. Rise time of the leading edge from the 10% point to the 90% point shall be 20 ns minimum, 50 ns maximum. Pulse rate shall be 576 kHz for both the 6 channel and 12 channel signals.

3.5.3.7.3 Squelch and alarm. The squelch shall disable the regenerated PCM signal when the signal-to-noise ratio at the unregenerated video output is less than 7 dB ± 2 dB when measured as in 3.4.3.5, with a reference deviation of ± 180 kHz peak. The audible and visible alarm shall operate in conjunction with the squelch to indicate locally, failure of the received signal from the remote station. A reversing switch shall be provided on the front panel such that regardless of the level of signal input to the receiver, the audible alarm can be disabled. The

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sensitivity of the squelch circuit shall be such that the point of operation which indicates a received signal (lamp extinguished) shall be -98 dBm ± 3 dB.

3.5.3.8 FDM demodulation requirements. With a signal deviated ± 75 kHz applied to the receiver antenna terminal, the level measured across the FDM output connector terminated in 91 ohms shall be .189V ± .02V rms. The meter of the receiver shall be in the green area with the switch in the FDM position.

3.5.3.9 Duplexers. Insertion loss at the center frequency of the pass-band of the transmitter section shall not exceed 2 dB over 90% of operating band and shall not exceed 2.5 dB over the remainder of the operating band. Insertion loss at the center frequency of the band-pass of the receiver section shall not exceed 3 dB.

3.5.3.10 Duplexer discrimination. The discrimination of the filter for the transmitter section shall be a minimum of 25 dB at ± 16 MHz and for the receiver section a minimum of 60 dB at ± 16 MHz and 100 dB at ± 50 MHz.

3.5.3.11 Voltage standing wave ratio. The voltage standing wave ratio measured at the antenna terminal of the duplexer, with the transmitter terminal terminated in a 50 ohm load and the receiver connected to the receiver connection, shall not exceed 2.2:1 at any transmitted frequency within the operating frequency range as specified in 3.5.

3.5.3.12 Systems level stability. The transmitter and receiver shall be connected through an attenuator with sufficient attenuation such that the power input to the receiver input terminals is minus 60 dBm. With the transmitter modulator as described in 3.5.3.4. (PCM modulation requirements) and the overall system adjusted for nominal gain, the following conditions shall be met: the system gain shall not vary more than +1 dB in any 8 hour period between the 91 ohm video input and 51 ohm video output terminals of the receiver when the received signal is varied between -95 dBm and -45 dBm under any combination of temperature, humidity or line voltage variation as specified herein.

3.5.4 Receiver Transmitter Order Wire.

3.5.4.1 Order wire operation. The order wire unit shall meet the following requirements to be compatible with Multiplexers TD-202 ( )/G, TD-204( )/G and TD-754( )/G.

3.5.4.1.1 Tone generator. When the tone generator is activated, the level measured at the TRANS “A” and TRANS “B” terminals of the order units shall be -10 dBm ± 1 dBm.

3.5.4.1.2 Signal receiver detector sensitivity. When a -10 dBm ± 6 dB, 1600 Hz ± 4 Hz signal is applied to either the REC A or REC B terminals of the order wire unit, the signal lamp shall light and the buzzer sound.

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3.5.4.2 Receiver demodulation order wire. With an RF signal deviated ± 45 kHz peak applied to the receiver antenna terminal, the level measured at the receiver order wire shall be -10 dBm ± 2 dB.

3.5.4.3 Transmitter deviation order wire. A 1000 Hz sinusoidal signal at a level of zero dBm applied to the microphone input terminals of the order wire unit shall produce a peak deviation of the transmitter between 38 kHz and 59 kHz. A 1000 Hz signal of -10 dBm applied to the transmitter order wire connector shall produce a peak deviation of between 39 kHz and 58 kHz.

3.5.4.4. Signal -to-noise order wire. The transmitter and receiver shall be connected together in a systems arrangement with sufficient attenuation between them such that the power measured at the input terminals of the receiver is -94 dBm. With the transmitter modulated as in 3.5.4.3 the signal-to-noise ratio measured with F1A weighting at the local order wire terminals (receive phone terminals) shall be at least 29 dB at any operating frequency in the presence of a normal 6/12 channel PCM signal and at least 40 dB in the FDM mode of operation. When measured with C-message weighing the signal-to-noise ratio shall be at least 27.5 dB and 38.5 dB respectively.

3.5.4.5 Peak limiting order wire. The limiting action to be provided in the transmitter for the order wire shall be such as to compress test tone power inserted at the microphone terminals in accordance with Table V.

TABLE V. Limiting Characteristics.

Input Power	Compression
-10 dBm	None
-5 dBm	None
0 dBm	Less than 0.5 dB
+ 5 dBm	3.5 dB ± 1 dB
+10 dBm	7.5 dB ± 2 dB
+15 dBm	10.0 dB ± 3 dB

Note: Measured between MIKE terminal and TRAN “A” or TRAN “B” Terminals.

3.5.4.6 Frequency response (order wire system). The overall frequency response characteristic between the local microphone input and the phone output terminals of the order wire unit at a distant station shall be plus 1 to minus 3 dB from 300 to 1800 Hz, referred to the 1 kHz response at an RMS level of zero dBm, when the equipment is employed in a full duplex PCM system.

3.5.4.7 Frequency response (order wire unit). The order wire unit RT-773( )/GRC-103 shall meet the following frequency response characteristics (Table VI) when measured with input levels specified in 3.5.4.11.

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TABLE VI. OW frequency response.

Path	From	To	Frequency Range	Response Ref: 1 kHz
"Thru"	REC "A"	TRAN "B"	200 Hz - 2.0 kHz	+0.5 dB - 1.0 dB
"Thru"	REC "B"	TRAN "A"	200 Hz - 2.0 kHz	+0.5 dB - 1.0 dB
Drop	REC "A"	PHONE terminal	200 Hz - 2.0 kHz	+0.5 dB - 1.5 dB
Drop	REC "B"	PHONE terminal	200 Hz - 2.0 kHz	+0.5 dB - 1.5 dB
Insert	MIKE Terminal	TRAN "A"	200 Hz - 2.0 kHz	+0.5 dB - 1.0 dB
Insert	MIKE Terminal	TRAN "B"	200 Hz - 2.0 kHz	+0.5 dB - 1.0 dB
Side Tone	MIKE Terminal	PHONE Terminal	200 Hz - 2.0 kHz	± 1.0 dB

3.5.4.8 Distortion (Order Wire System). With the equipment arranged as in 3.5.4.4, "Signal-to-Noise", the total harmonic distortion measured at the order wire output terminals shall not exceed 5%.

3.5.4.9 Distortion (Order Wire Units). The distortion of the order wire unit when measured with a 1000 Hz signal as specified in 3.5.4.7 shall be as follows:

"THRU" PATHS	REC "A" to TRAN "B"	Less than 1%
	REC "B" to TRAN "A"	Less than 1%
INSERT PATHS	MIKE terminal to TRAN "A"	Less than 2%
	MIKE terminal to TRAN "B"	Less than 2%
DROP PATHS	REC "A" to PHONE terminal	Less than 1%
	REC "B" to PHONE terminal	Less than 1%

3.5.4.10 Net loss (order wire system). The net loss between the local microphone input and the phone output at a distant station shall be 18 dB ± 3 dB.

3.5.4.11 Order wire unit levels. The order wire unit levels when measured with a 1000 Hz signal shall conform to Table VII.

TABLE VII. Unit Levels (Order Wire).

From	To	Output Level	Input Levels
REC "A"	TRAN "B"	-10 dBm ± 0.1 dB (adjustable)	-10 dBm
REC "B"	TRAN "A"	-10 dBm ± 0.1 dB (adjustable)	-10 dBm
REC "A"	Phone Terminal	-18 dBm ± 1.0 dB	-10 dBm

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REC “B”	Phone Terminal	-18 dBm ± 1.0 dB	-10 dBm
MIKE Terminal	TRAN “A”	-10 dBm ± 1.0 dB	0.dBm
MIKE Terminal	TRAN “B”	-10 dBm ± 1.0 dB	0 dBm
MIKE Terminal	Phone Terminal	-24 dBm ± 2.0 dB (side tone)	-10 dBm
REC “A”	TRAN “A”	-55 dBm ± 10.0 dB (crosstalk)	-10 dBm
REC “B”	TRAN “B”	-55 dBm ± 10.0 dB (crosstalk)	-10 dBm

3.5.5 Antenna.

3.5.5.1 Symmetry. The ratio of the half-power (-3 dB) E- and H- plane beam width shall not exceed 2.0:1 over the frequency range.

3.5.5.2 Squint. The gain of the antenna measured in the direction of the normal from the antenna shall not differ from the maximum gain measured at any frequency in the operating range by more than 0.5 dB.

3.5.5.3 Minor-Lobes. The ratio of the major lobe of the antenna to any minor lobe shall be at least 12 dB at any frequency in the operating range.

3.5.5.4 Gain. The gain of the antenna relative to an isotopic radiator shall be at least 13.40 dB at 695 MHz, 13.65 dB at 750 MHz, 13.85 dB at 800 MHz, 14.05 dB at 850 MHz, 14.30 dB at 900 MHz, 14.50 at 950 MHz and 14.70 dB at 1000 MHz.

3.5.5.5 Voltage standing wave ratio. The VSWR measured at the antenna terminals shall not exceed 1.75:1 over the operating frequency range.

3.5.5.6 Impedance. The nominal input impedance of each antenna shall be 50.0 ohms.

3.5.6 Mast.

3.5.6.1 Erection. The mast shall be designed to be manually erected in the field with antenna and transmission lines installed using common hand tools. Erection shall be facilitated in wind conditions of up to 25 mph with launcher guyed and all other wires slack.

3.5.6.2 Operation. The mast assembly, erected with guy wires tensioned and antenna and transmission line installed, shall withstand the following conditions without permanent deformation and maintain the antenna azimuth ± 5° of any initial setting.

a. Wind velocities up to 75 mph; or

b. Wind velocities up to 50 mph with a one-half inch radial ice load on the entire mast-antenna installation.

3.5.7 Radio Set component interchangeability. All the components, subassemblies, and replaceable parts comprising the Radio Set AN/GRC-103(V)3 shall be physically and

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functionally interchangeable without modification or adjustment. All of these components, subassemblies and replaceable parts shall be examined prior to their final assembly to ensure that they conform to the physical and functional requirements contained in their respective drawings. The following paragraphs (3.5.7.1 through 3.5.7.33) pertain to the requirements of these items:

3.5.7.1 Receiver, Radio R-1329 ( )(P)/GRC-103(V). The receiver, radio shall meet the following requirements:

3.5.7.1.1 With an RF input at a frequency of 30 MHz (FM modulated) and a level of -30 dBm to -70 dBm into a 50 ohm ± 10% impedance, the FM demodulation shall be as follows:

Video	1.25 mV/KHz ± 10% into 50 ohms
FDM	3.6 mV/KHz ± 10% into 91 ohms
Order Wire	7.7 mV/KHz ± 10% into 600 ohms

3.5.7.1.2 The video Frequency response shall be Gaussian, with 3 dB attenuation at 275 KHz with reference to attenuation at 10 KHz.

3.5.7.1.3 Outputs shall be as follows:

Regenerated PCM	-2 v ± 10% peak on 91 ohms
Timing	2 v ± 10% Amplitude on 91 ohms
Video and FDM	See Demodulation

3.5.7.1.4 Output return loss shall be as follows:

Video output (50 ohms) return loss shall be 26 dB minimum
FDM output (91 ohms) return loss shall be 26 dB minimum

3.5.7.1.5 RF output (from doubler or local oscillator) shall be as follows:

Frequency range	95 MHz to 145 MHz
Output frequency selection	in steps of 62.5 KHz or 41.666 KHz
Output power	+ 14 dBm minimum
Output impedance	50 ohms nominal

3.5.7.2 Amplifier-converter AM-4318/GRC-103(V). The amplifier-converter shall meet the following requirements:

- The high signal alarm shall operate at levels above + 15 dBm and cease operation at levels below -15 dBm at frequencies between 695 and 1000 MHz.
- The noise figure shall be 10 dB maximum from 695 to 1000 MHz.
- The input VSWR ref. 50 ohms shall be 3.5:1 maximum from 690 to 1000 MHz.
- The IF output at 30 MHz shall be -29 dBm maximum to -60 dBm minimum for -80 dBm RF input with AGC voltage between -9.9 and -10.8 V

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- The local oscillator output shall be + 6 dBm minimum at 725 to 1030 MHz for inputs of +15 dBm ± 2 dB at 95 to 145 MHz.
- The meter indication shall be 0.02 mA minimum into 5000 ohms for lo level of + 6 dBm

3.5.7.3 Transmitter, Radio T-983( )(P)/GRC-103(V). The transmitter, radio shall meet the following requirements:

With a video input of 1 to 4 volts peak-to-peak at 125 to 960 Hz into an impedance of 91 ohms ± 10%, the RF output over a frequency rang of 93.666 to 144.875 MHz ± .001% shall be 3 w +2 dB -1 dB into a 50 ohm impedance. The frequency deviation shall be up to 150 KHz peak.

3.5.7.4 Amplifier-frequency Multiplier AM-4322/GRC-103(V). The amplifier frequency multiplier shall meet the following requirements:

With an input of 2 watts minimum at a frequency range of 95 to 145 MHz, the output power at the frequency range of 695 to 1000 MHz (Channel 990 to Channel 1600) shall be 20 watts (+43 dBm) minimum for all channels.

3.5.7.5 Synthesizer (1RE1A2/5TR1A2). The synthesizer shall meet the following requirements:

Frequency Range	47.5 - 72.5 MHz
Frequency Accuracy	± 250 Hz @ 47.5 MHz ± 300 Hz @ 60.234375 MHz ± 400 Hz @ 72.5 MHz
Output power (47.5 - 72.5 MHz)	+ 14 dBm ± 1.5 dB
Modulation Sensitivity	265KHz/Volt ± 0.75 dB
Meter output	50 µA ± 30 µA

3.5.7.6 Restorer, Pulse Form (1RE1A3). The restorer, pulse form shall meet the following requirements:

3.5.7.6.1 The regenerated PCM video output on Pin 13 of connector P1 shall be -2 volts peak ± 10 percent into 91 ohms.

3.5.7.6.2 The timing pulses phase locked to the regenerated PCM video rate on Pin 15 of connector P1 shall be 2 volts peak ± 10 percent into 91 ohms.

3.5.7.6.3 The orderwire audio on Pin 14 of connector P1 shall be -10dBm ± 1 dB into 400 ohms.

3.5.7.6.4 The PCM metering current on Pin 9 of connector P1 shall be 40 to 60 µA into 5000 ohms.

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3.5.7.6.5 The order wire metering current on Pin 10 of connector P1 shall be 30 to 70 μA int 5000 ohms.

3.5.7.7 Limiter-Discriminator (1RE1A4). The limiter-discriminator shall meet the following requirements:

With an IF signal on Connector P1 at a center frequency of 30 MHz and a level of -4 dBm nominal (-30 dBm to 0 dBm) and a peak deviation of ± 300 KHz, the output baseband signal on Connector P2 shall meet the following over the range of 125 Hz to 960 KHz:

Frequency Range	125 Hz to 960 KHz
Demodulation Sensitivity	0.5 v ± 3% per MHz
Phase of Demodulation	Positive going with increasing frequency
Demodulation Linearity	Maximum Variation of derivative 3% ± 600 KHz 10% ±1000 KHz
Demodulation Linearity Offset	Decreasing 6% from lower to higher frequency
Frequency Response	125 Hz to 960 KHz; - 0.5 dB to + 1.5 dB (Ref. 10 KHz)

3.5.7.8 Panel Monitor (1RE1A1). The panel, monitor shall be examined for continuity.

3.5.7.9 Power Supply (1RE1PS1). The power supply shall meet the following requirements:

3.5.7.9.1 Output Voltages. With input of 115 Vac RMS ± 10% at a frequency of 50 Hz to 400 Hz ± 5%, the output voltages shall be as below:

Required Output	Tolerance	Maximum Current	Ripple
24 Vdc Regulated Floating	±0.1 v	1.0 A	2 mv p-p 120Hz Max
+ 12 Vdc Regulated	±0.1 v	1.0 A **	
- 12 Vdc Regulated	±0.1 v	1.0 A **	
+ 26 Vdc Unregulated	± 1 v	3 A for 15 sec	600mv p-p 120 Hz at nom current 500 mA

** To operate at the maximum current rating of one of the 12 v sources a comparative amount of current must be drawn from the other 12 v source.

3.5.7.9.2 Regulation. The regulation of the + 24 v, +12 v, and -12 v supplies shall be less than ± 0.5 v for an input voltage of 115 Vac ±10%, 60 Hz over the temperature range of -40°C to +80°C and a variable load current of 0 - 1 A.

3.5.7.10 Amplifier-Frequency Multiplier (1RE1A5). The amplifier-frequency multiplier shall meet the following requirements:

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With inputs at a level of + 14 dBm ± 1.5 dB at frequencies from 47.5 to 72.5 MHz and VSWR less than 2:1 on Pin A1 of Connector P1, tune for second harmonic output of + 14 dBm +3 dB -0 dB. The harmonic rejection shall be at least 40 dB at f_0 of 60 MHz and at least 20 dB at 3 f_0 and 4 f_0 measured at Pin A2 of Connector P1.

3.5.7.11 Amplifier, Video (1RE1AR1). The amplifier, video shall meet the following requirements:

3.5.7.11.1 Outputs. The video output impedance shall be 56 ohms ± 2%. The video amplification factor shall be 18.6 ± 2%, adjusted unloaded, and shall be adjustable between 10 and 50 when video output is terminated in 51 ohms. The PCM output impedance shall be 1000 ohms ± 5%. the PCM amplification factor shall be adjustable between 20 and 100 when PCM output is unloaded.

3.5.7.11.2 Frequency response. The Gaussian response shall be as indicated below:

Frequency KHz	Attenuation dB
.125	1 ± 1
.200	0 ± .6
.250	0 ± .5
1.0	0 ± .1
10.0	Reference
205.	.75 ± .2
410.	3.0 ± .3
615.	6.65 ± .6
820.	12.0 ± .7
960.	16.5 ± 1.7

3.5.7.11.3 Meter sensitivity. For 24 channel PCM, the meter sensitivity shall be set for a 50 % indication for a 1.98 volt root mean square value output (Video out unloaded) on a 100 µA full scale deflection 5K resistance meter. For FDM, , the meter sensitivity shall be set for a 50 % indication for a 0.50 volt root mean square value output (Video out unloaded) on a 100 µA full scale deflection 5K resistance meter.

3.5.7.11.4 Low signal alarm. The low signal alarm shall operate with a simulated output of 25 mv at 600 KHz.

3.5.7.12 Amplifier, Intermediate (1RE1AR2). The amplifier, intermediate shall meet the following requirements:

At -10 dBm output the gain shall be 50 dB ± 5 dB
At -40 dBm input the gain shall be 37 dB ±1.5 dB.
At -10 dBm output the variation in gain between 29 and 31 MHz shall be ±1.5 dB max.

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3.5.7.13 Distribution Box (1A2A1). The distribution box shall be examined for continuity.

3.5.7.14 Tuner, Radio Frequency (34A1). The tuner shall meet the following requirements:

3.5.7.14.1 High Signal Alarm. The high signal alarm shall operate at signal levels above + 15 dBm and shall cease operation at levels below -10 dBm.

3.5.7.14.2 Noise. The noise figure shall be 10 dB max at frequencies from 695 to 1000 MHz.

3.5.7.14.3 Input Match. The Input VSWR shall be less than 3.5:1 ref. 50 ohms from 695 to 1000 MHz.

3.5.7.14.4 IF Output. With a -80 dBm RF input, the IF output at 30 MHz shall be between -29 dBm and -60 dBm with AGC voltage between -9.9 and -10.8 volts.

3.5.7.14.5 Local Oscillator Drive. With input of +15 dBm ± 2 dB at 95 to 145 MHz, the output shall be +6 dBm minimum at 725 to 1030 MHz.

3.5.7.14.6 Multimeter indication. For a local oscillator level of +6 dBm, the multimeter shall indicate 0.02 mA minimum into 5000 ohms.

3.5.7.15 Duplexer (34A1A1). The duplexer shall meet the following requirements:

Parameter	Requirements
VSWR at antenna port - ref 50 ohms	.
Transmitter Frequency ± 1.0 Mhz	2.0:1 maximum
Receiver Frequency ± 1.o Mhz	2.0:1 maximum
Transmitter to Receiver Frequency Separation	16 MHz minimum
Tuning range	695.0 MHz to 1000 MHz (channels 990 to 1600)

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Error of channel indication	± 0.5 in from normal
Insertion loss at center of pass band	
Transmitter section	2.0 dB maximum over90% of tuning range 2.5 dB maximum over remainder
Receiver section	3.0 dB maximum
Variation of Loss within Pass Band	
Transmitter section	1.0 dB maximum within ±1.0 Mhz
Receiver section	1.0 dB maximum within ±1.0 Mhz
Discrimination	
Transmitter section	25 dB minimum at ± 16 MHz
Receiver section	60 dB minimum at ± 16 MHz 100 dB minimum at ± 50 MHz

3.5.7.16 Converter, Frequency, Electronic (34A2). The converter, frequency, electronic shall meet the following requirements:

With an input of +15 dBm ±2 dB into a VSWR of 1.75:1 Max ref 50 ohms at a frequency range of 95 to 145 Mhz, The output shall be between +6.0 dBm and +14.0 dBm into a VSWR mismatch of 2:1 at a frequency of 725 to 1030 MHz.

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3.5.7.16.1 The frequency multiplier assembly shall meet the following requirements:

Parameter	Requirement
Input	
Frequency rang	170 to 290 MHz
Level	+ 17.0 dBm ± 2.0 dB
VSWR Ref 50 ohms	1.75:1 Max.
Output	
Frequency range	725 to 1030 MHz
Level into 2:1 VSWR mismatch ref 50 ohms	+7.3 dBm min. 725 to 840 MHz +8.3 dBm Min. 840 to 1030 MHz
3 dB bandwidth	0.8% min. of output Frequency 3.0% max. of output frequency

3.5.7.16.2 The Filter, Bandpass assembly shall meet the following requirements:

Parameter	Requirement
Frequency range	725 to 1030 MHz
Insertion loss ref; 50 ohms	2 dB max.
3 dB bandwidth	5 to 10 MHz
Rejection at f_0 ± 50 MHz	35 dB min.
VSWR; Ref 50 ohms	1.6:1 Max.
Tuning Law	See Table

Tuning Law

Center Frequency f_0 Mhz	Plunger Travel inches	Tolerance MHz
725	0.0	± 1
800	0.370 ± 0.004	± 1
880	0.812 ± 0.004	± 1
950	1.144 ± 0.004	± 1
1030	1.496 ± 0.004	± 1

3.5.7.16.3 The assembly shall meet the following requirements:

Parameter	Requirement
Input	
Frequency range - f_0	95 to 145 MHz
Level	+ 15.0 dBm ± 2.0 dB
VSWR Ref 50 ohms	1.75:1 Max.
Output	
Frequency range - $2f_0$	190 to 290 MHz
Level into 2:1 VSWR mismatch ref 50 ohms	+17. dBm ± 2.5 dB
Rejection of spurious responses up to 4 f_0	20 dB minimum

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3.5.7.17 Mixer Stage, Frequency (34A7). The mixer stage, frequency shall meet the following requirements:

Parameter	Requirement
Power Requirements	+ 12 v, 15 mA max. - 12 v, 5 mA max.
Frequency of operation	695 MHz to 1000 MHz
LO Frequency	725 MHz to 1030 MHz
Output frequency (IF)	30 MHz
Input VSWR	2.0:1 max. ref 50 ohms
LO level	+ 6 dBm to + 14 dBm
Noise figure	3.6 dB max. with 3.0 dB IF AMP noise figure
Rejection at f_0 + 60 MHz	15 dB below wanted signal
Conversion gain	14 dB ± 4 dB
LO level metering	135 mV min. across 5 k ohms for + 6 dBm LO input

3.5.7.18 Control Indicator Coding Check (34A4). The control indicator shall meet the requirements for coding of channel numbers.

3.5.7.19 Control Monitor, Signal Level (34A5). The control monitor shall meet the following requirements:

At frequencies from 395 MHz to 1000 MHz with less than -20 dBm applied power, the input VSWR shall be less than 1.3:1 and the insertion loss shall be less than 0.75 dB. The isolation shall be 28.5 dB minimum at + 45.5 dBm input. The high signal alarm operate level shall be between -10 dBm and + 10 dBm.

3.5.7.20 Panel, Monitor (5TR1A1). The panel monitor shall be examined for continuity.

3.5.7.21 Control, Alarm (5TR1A3). The control alarm shall meet the following requirements:

3.5.7.21.1 Connections.

Connector	Pin	Function	Voltage	Measured to pin	Current maximum
P2	1	Input +	0.5 v Maximum	2, 3, 6	-
P2	2, 3, 6	Input Com	Gnd	-	-
P2	7, 9	Supply	26 VAC ±2 v 400 Hz	8	50 mA
P2	8	Supply Com	-	-	-
P2	4	Alarm	50 VDC Maximum	5	50 mA
P2	5	Alarm Com	-	-	-

3.5.7.21.2 Input Impedance. The input impedance shall be 100 Kilohms maximum.

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3.5.7.21.3 Alarm Operating Point. The alarm operating point shall be potentiometer set with a range of setting from 75 mv maximum to 160 mv minimum.

3.5.7.21.4 Alarm Extinguish Backlash. When the alarm is set to operate at 80 mv ± 5 mv, the alarm extinguish point shall be between + 110 and + 150 mv.

3.5.7.21.5 Alarm Output. The contact shall be closed in the alarm position.

3.5.7.22 Amplifier-Frequency Multiplier (5TR1A4). The amplifier-frequency multiplier shall meet the following requirements:

With inputs at a level of + 14 dBm ± 1.5 dB at frequencies from 47.5 to 72.5 MHz and VSWR less than 2:1 on Pin A1 of Connector P13, tune for second harmonic output of + 34 dBm minimum. The harmonic rejection shall be at least 40 dB at f_0 and at 3 f_0 and at least 35 dB at 4 f_0 measured at Pin A2 of connector P13.

3.5.7.23 Amplifier-Monitor (5TR1A5). The amplifier-monitor shall meet the following requirements:

- The video signal output level shall be 0.3 v maximum peak into 500 ohms.
- The video voltage gain shall be -0 dB ±1.0 dB into 500 ohms attenuated by 0,3.5, 6, 9.5 or 12 dB.
- The order wire voltage gain shall be -17.7 dB ± 0.2 dB into 500 ohms.

With 0 dB video attenuation, the following outputs shall correspond to +50 µA ± 2% dc current into 5000 ohms at the corresponding metering output:

24 channel	0.3 v peak
12 channel	0.16 v peak
FDM	0.075 peak

The video frequency gain at higher frequencies relative to the gain at 10 Khz shall be within the limits below:

Frequency KHz	Rel Gain dB	Limits
283	-0.75	± 0.2
567	-3.0	± 0.2
850	-6.75	± 0.3
960	-8.6	± 0.6

3.5.7.24 Power Supply (5TR1PS1). The power supply shall meet the following requirements:

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3.5.7.24.1 Output Voltages. With input of 115 Vac RMS ± 10% at a frequency of 50 Hz to 400 Hz ± 5%, the output voltages shall be as below:

HT Supply (to Ground)	+630 Vdc ± 32 v, Unregulated, at 200 mA, 20 v p-p maximum ripple.
Relay and Lamp Supply (Floating)	26 Vdc ± 1.3 v, Unregulated, at 500 mA, 1 v p-p maximum ripple
Filament Supplies, (2) (Floating)	6.8 Vdc + 0.7v -0 v, Unregulated, at 1.3 A, 250mv maximum ripple
28 v Supply (to Ground)	+ 28.0 Vdc, Regulated, nominally 1.0 A, 15mv p-p maximum ripple
12 v Supply (to Ground)	+ 12.0 Vdc, Regulated, nominally 250 mA, 10 mv p-p maximum
Timer Supply (P/A) (to Ground)	+ 6.2 Vdc ±0.3 v, nominally, 10 µA.
Cooler Supply (Floating)	128 v peak ± 6 v, Unregulated, 400 Hz ± 20 Hz square wave, 80 VA Maximum output, 10 Ω source impedance.
Low Power Alarm Supply (Floating)	52 v peak ± 5.0 v, Unregulated, 400 Hz ± 20 Hz, Square Wave, Center tapped, 2 W maximum output.
Constant Current Sources	60 mA, voltage range 6.5 v to 150 v maximum, 160 mA, voltage range 3.5 v to 150 v maximum.

3.5.7.24.2 Special Features. The power supply shall be equipped with the following special features:

HT Trip Feature	Removes HT output.
Overload Protection	Circuit limiting or short circuit protection on all outputs.
Elapsed time Indicator	0 to 10,000 hours.

3.5.7.25 Distribution Box (5A2A1). The distribution box shall be examined for continuity.

3.5.7.26 Amplifier, Radio (38AR1). The amplifier, radio shall meet the following requirements:

Parameter	Requirement
Input Frequency range	 695 to 1000 MHz

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Power level for specified min. output power	+ 21 dBm max.(130 mW)
VSWR; ref 50 ohms	2.0:1 max.

Parameter	Requirement
Output	
Frequency range	695 to 1000 MHz
Power level	+ 43 dBm Min. (20 w)
Impedance	50 ohms nominal
General Specifications	
Error of tuning indication	±0.5 inch max. from nominal indication
Automatic level control	1 mA min. into short circuit for 15vdc on output tube cathode
Power requirements	630 Vdc 200 mA max. 7.2 Vdc 1.5 a max. (2 required) constant current supply 145 mA nominal constant current supply 62 mA nominal
Metering For Specified Min. Output Power	
Power output	0.044 mA ± 0.010 mA
Reflected power	0.065 mA ± 0.010 mA
Driver power	0.050 mA ± 0.010 mA

3.5.7.27 Frequency Multiplier Assembly (38A1). The frequency multiplier assembly shall meet the following requirements with input frequencies from 105 to 140 Mhz at a nominal level of +34 dBm and a VSWR of 2.0:1 max. ref. 50 ohms:

Parameter	Requirement
Output Frequency	695 to 1000 MHz
Output Level	+21 dBm min. into2.0:1 VSWR
Output level control for 17 dBm output	+3.0 v Max.
Output level control for maximum output	0 v Max.
3 dB bandwidth	13.5 MHz ±8.5 MHz
Output level metering at + 21 dBm output	250 mV ± 50 mV across 5000 ohms shunted by 1N277 diode

Tracking requirement shall be in accordance with the Table below:

Input freq. MHz	Output Freq.	Output Power	Counter	Adjustment Control

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		MHz		Reading	
Test Set	AUT				
62.50	125.00	1000	+ 22 dBm	0000	C2, C3
61.25	122.50	980	min	0009.5	
60.00	120.00	960		0020	
58.75	117.50	940		0030.5	
57.50	115.00	920		0040.5	C1 No. 1, C4 No. 1
56.25	112.50	900		0051.5	
Input Freq. MHz		Output Freq. MHz	Output Power	Counter Reading	Adjustment Control
Test Set	AUT				
55.00	110.00	880	+ 22 dBm	0062.5	
53.75	107.50	860	min	0074.5	
52.50	105.00	840		0089	C1 No. 2, C4 No. 2
68.34	136.67	820		0103	
66.67	133.33	800		0117	
65.00	130.00	780		0134.5	C1 No. 3, C4 No. 3
63.34	126.67	760		0151.5	
61.67	123.33	740		0168	
60.00	120.00	720		0185.5	
58.33	116.67	700		0207.5	
57.92	115.83	695		0214.5	C1 No. 4, C4 No. 4

3.5.7.28 Control-Indicator (38A2). The control-indicator shall meet the requirements for indication of channel numbers.

3.5.7.29 Panel, Distribution (9A1). The panel, distribution shall meet the following resistance requirements:

From		To		
Connector	Pin	Connector	Pin	Resistance, Ohms
P1	6	J2	A	0
P1	3	Buzzer	Negative Lead	0

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J1	A	J1	B	300 ± 60
TB1	1	TB1	2	Less than 2000
J4	A	J4	J	50 ± 5
J4	A	J4	C	Actuate Press-to-talk Less than 2000
J2	K	J2	G	620 ± 60
J1	K	J1	G	620 ± 60

3.5.7.30 Panel, Monitor (9A2). The panel monitor shall meet the following resistance requirements:

Between Pins	Switch	Position	Resistance ohms
10 and 11	S1	Normal	0
7 and 8	S1	Normal	0
11 and 12	S1	Pressed	0
8 and 9	S1	Pressed	0
10 and 11	S1	Pressed	∞
7 and 8	S1	Pressed	∞

Between Pins	Switch	Position	Resistance ohms
3 and 5	S2	On	0
5 and 6	S2	On	200 ± 20%
3 and 5	S2	Off	∞
5 and 6	S2	Off	400 ± 20%
6 and 13	-	-	200 ± 20%
1 and Ground	-	-	0
2 and Ground	-	-	∞
4 and Ground	-	-	∞
14 and Ground	-	-	∞
15 and Ground	-	-	∞

3.5.7.31 Converter, Telephone Signal (9A3). The converter, telephone signal shall meet the following requirements:

With an input signal of 200 mv nominal at a frequency of 1600 Hz ± 4 Hz on Pin L of connector P1, the output on Pin F of connector P1 shall be 1.6 volts ± 0.1 volt at a frequency of 1600 Hz ± 4 Hz into a 600 ohm termination.

3.5.7.32 Amplifier Assembly (9A4). The amplifier assembly shall meet the following requirements:

3.5.7.32.1 Power Gain.

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From	To	Power Gain	Nominal Input Levels
Rec. “A”	Tran. “B”	0 dB ± 0.1 dB (adjustable)	- 10 dBm
Rec. “B”	Tran. “A”	0 dB ± 0.1 dB (adjustable)	- 10 dBm
Rec. “A”	Phone Terminal	-8 dB ± 1.0 dB	- 10 dBm
Rec. “B”	Phone Terminal	-8 dB ± 1.0 dB	- 10 dBm
Mike Terminal	Tran. “A”	-10 dB ± 1.0 dB	0 dBm
Mike Terminal	Tran. “B”	-10 dB ± 1.0 dB	0 dBm
Mike Terminal	Phone Terminal	-24 dB ± 3.0 dB (Sidetone)	0 dBm
Rec. “A”	Tran. “A”	-45 dB ± 10.0 dB (Crosstalk)	- 10 dBm
Rec. “B”	Tran. “B”	-45 dB ± 10.0 dB (Crosstalk)	- 10 dBm

3.5.7.32.2 Limiting Characteristics, Insertion Path. measured between Mike terminal and Tran A or Tran B terminals

Input Power	Compression
-10 dBm	None
-5 dBm	None
0 dBm	Less than 0.4 dB

Input Power	Compression
+5 dBm	3.5 dB ± 1 dB
+10 dBm	7.5 dB +2dB -1 dB
+15 dBm	10.0 dB +3 dB -1 dB

3.5.7.32.3 Frequency Response.

Path	From	To	Frequency Range	Response Ref: 1 KHz
Thru	Rec A	Trans B	200 Hz - 2 KHz	+0.5 dB, -1.0 dB
Thru	Rec B	Trans A	200 Hz - 2 KHz	+0.5 dB, -1.0 dB
Dropping	Rec B	Phone Term	200 Hz - 1 KHz	0.0 dB -1.0 dB
			1 KHz - 2 KHz	+0.5 dB -0.0 dB
Dropping	Rec A	Phone Term	200 Hz - 1 KHz	0.0 dB -1.0 dB
			1 Khz - 2 KHz	+0.5 dB -0.0 dB

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Insertion	Mike Term	Trans A	200 Hz - 1 KHz 1 KHz - 2 KHz	+0.5 dB 0.0 dB 0.0 dB -1.0 dB
Insertion	Mike Term	Trans B	200 Hz - 1 KHz 1 KHz - 2 KHz	+0.5 dB 0.0 dB 0.0 dB -1.0 dB
Sidetone	Mike Term	Phone Term	200 Hz - 2 KHz	±1.0 dB

3.5.7.32.4 Harmonic Distortion. The harmonic distortion on the "Thru" paths shall be less than 1%. The harmonic distortion on the insertion paths shall be less than 2 %. The harmonic distortion on the dropping paths shall be less than 1%.

3.5.7.33 Dummy Load (38AR1AT1). The dummy load shall meet the following requirements:

The VSWR shall be less than 1.3:1 ref 50 ohms over the frequency range of 695 MHz to 1000 MHz. The VSWR shall be less than 1.9:1 ref 50 ohms over the frequency range of 1350 MHz to 1850 MHz.

3.5.7.34 Preconditioning. The above components, assemblies, subassemblies, and replaceable parts shall be capable of meeting the requirements herein without subsequent processing after being subjected to the preconditioning of 4.6.

3.6 Environmental conditions.

3.6.1 Temperature.

3.6.1.1 Low temperature. The radio set shall not be damaged or suffer no degradation of performance when exposed to storage temperatures of -65°F while non-operating, and operation at temperature of -35°F. The storage temperature shall be -65°F and the operating temperature shall be -35°F. Prior to and following low temperature exposure, the equipment shall meet full specification performance for the following measurements:

a. Transmitter frequency accuracy of better than 0.002% and power output of 20 watts minimum except the test shall be performed at channels 990, 1100, 1210, 1340, 1470, and 1600. Tests at line frequencies of 47.5 Hz and 420 Hz and high and low line voltage shall be performed at any one channel

b. Transmitter power to the antenna of 13 watts minimum with the tests performed at channels 990, 1360, and 1600.

c. Transmitter frequency response in accordance with Table I of Paragraph 3.5.1.5 and receiver noise figure in accordance with Paragraph 3.5.2.2 except tests be performed at channels 900, 1360, and 1600.

d. Receiver noise figure shall not exceed 10 dB with the test performed at channel 1400.

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e. All indicator lights, meters, and alarms shall operate as specified.

f. Overall frequency response in accordance with Table III of Paragraph 3.5.3.2, except frequencies below 10 kHz be deleted from frequency response Measurement.

g. Rise time of the PCM pulse from 10% to 90% amplitude shall be between 10 ns and 50 ns. Decay time of the PCM pulse from 90% to 10% amplitude shall be between 20 ns and100 ns. PCM pulse width at 50% amplitude shall be between 90 ns and 150 ns. The error rate shall be less than 5 errors/second at a received carrier level of -92 dBm.

h. Paragraph 3.5.4.4.

i. The receiver protection switch shall function as specified.

3.6.1.2 High temperature. The radio set shall not be damaged or suffer any degradation of performance when exposed to a temperature up to 125°F. Prior to and following high temperature exposure, the equipment shall meet full specification performance for the following measurements:

a. Transmitter frequency accuracy of better than 0.002% and power output of 20 watts minimum except the test shall be performed at channels 990, 1100, 1210, 1340, 1470, and 1600. Tests at line frequencies of 47.5 Hz and 420 Hz and high and low line voltage shall be performed at any one channel

b. Transmitter power to the antenna of 13 watts minimum with the tests performed at channels 990, 1360, and 1600.

c. Transmitter frequency response in accordance with Table I of Paragraph 3.5.1.5 and receiver noise figure in accordance with Paragraph 3.5.2.2 except tests be performed at channels 900, 1360, and 1600.

d. Receiver noise figure shall not exceed 10 dB with the test performed at channel 1400.

e. All indicator lights, meters, and alarms shall operate as specified.

f. Overall frequency response in accordance with Table III of Paragraph 3.5.3.2, except frequencies below 10 kHz be deleted from frequency response Measurement.

g. Rise time of the PCM pulse from 10% to 90% amplitude shall be between 10 ns and 50 ns. Decay time of the PCM pulse from 90% to 10% amplitude shall be between 20 ns and100 ns. PCM pulse width at 50% amplitude shall be between 90 ns and 150 ns. The error rate shall be less than 5 errors/second at a received carrier level of -92 dBm.

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h. Paragraph 3.5.4.4.

i. The receiver protection switch shall function as specified.

3.6.2 Humidity (moisture-resistance) test. The radio set shall not be damaged or suffer any degradation of performance when exposed to relative humidities of 95%. During and after such exposure, the equipment shall meet full specification performance for the measurements below:

a. Transmitter frequency accuracy of better than 0.002% and power output of 20 watts minimum except the test shall be performed at channels 990, 1100, 1210, 1340, 1470, and 1600. Tests at line frequencies of 47.5 Hz and 420 Hz and high and low line voltage shall be performed at any one channel

b. Transmitter power to the antenna of 13 watts minimum with the tests performed at channels 990, 1360, and 1600.

c. Transmitter frequency response in accordance with Table I of Paragraph 3.5.1.5 and receiver noise figure in accordance with Paragraph 3.5.2.2 except tests be performed at channels 900, 1360, and 1600.

d. Receiver noise figure shall not exceed 10 dB with the test performed at channel 1400.

e. All indicator lights, meters, and alarms shall operate as specified.

f. Overall frequency response in accordance with Table III of Paragraph 3.5.3.2, except frequencies below 10 kHz be deleted from frequency response Measurement.

g. Rise time of the PCM pulse from 10% to 90% amplitude shall be between 10 ns and 50 ns. Decay time of the PCM pulse from 90% to 10% amplitude shall be between 20 ns and100 ns. PCM pulse width at 50% amplitude shall be between 90 ns and 150 ns. The error rate shall be less than 5 errors/second at a received carrier level of -92 dBm.

h. Paragraph 3.5.4.4.

i. The receiver protection switch shall function as specified.

NOTE: Measurements shall be made during the periods shown on the figure for the cycle. Do not move equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If any equipment fails to meet the specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment still fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the

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equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test.

3.6.3 Altitude (elevation) test. The radio set shall not be damaged or suffer degradation of performance when subjected to altitudes of 25,000 feet above sea level or 11.1 inches of Hg. During and after exposure to 25,000 feet in altitude, the equipment shall meet full specification performance for the measurements specified below. The chamber may be returned to ambient atmospheric pressure for frequency changes.

a. Transmitter frequency accuracy of better than 0.002% and power output of 20 watts minimum except the test shall be performed at channels 990, 1100, 1210, 1340, 1470, and 1600. Tests at line frequencies of 47.5 Hz and 420 Hz and high and low line voltage shall be performed at any one channel

c. Transmitter frequency response in accordance with Table I of Paragraph 3.5.1.5 and receiver noise figure in accordance with Paragraph 3.5.2.2 except tests be performed at channels 900, 1360, and 1600.

d. Receiver noise figure shall not exceed 10 dB with the test performed at channel 1400.

g. Rise time of the PCM pulse from 10% to 90% amplitude shall be between 10 ns and 50 ns. Decay time of the PCM pulse from 90% to 10% amplitude shall be between 20 ns and100 ns. PCM pulse width at 50% amplitude shall be between 90 ns and 150 ns. The error rate shall be less than 5 errors/second at a received carrier level of -92 dBm.

3.6.4. Vibration and shock.

3.6.4.1 Vibration/Bounce (class L units). The radio set shall not be damaged or suffer any degradation of performance when subjected to simple motion having an amplitude of ± 0.015 inch (0.030 inch total excursion) with a frequency being varied between 5 and 55 Hz.

3.6.4.2 High intensity shock. The radio set shall not be damaged or suffer any degradation of performance when subjected to a sequence of shocks applied in each direction along each of two mutually perpendicular axis (horizontal and vertical). The shocks shall be half sine pulses having a peak amplitude of 70 g's ± 10g's at a time duration of 4 milliseconds ±5%.

3.6.4.3 Shock, bench-handling. The radio set shall not be damaged or suffer any degradation of performance following shocks resulting from bench-handling. After being subjected to shocks, the equipment shall meet full specification performance for the measurements specified in Table VIII.

TABLE VIII. Bounce and shock.

Paragraph heading	Requirement	

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	paragraph	Performance after test
Vibration/Bounce (class L units)	3.6.4.1	Specified performance. (notes A and B). No physical damage.
Shock high intensity	3.6.4.2	Operable (Note B). Any physical damage shall be minor only.
Shock; bench-handling	3.6.4.3	Specified performance (Note B). No physical damage

NOTE A: With the radio set shock-mounts (if any) blocked, the amplitude of vibration of any part, subassembly, or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 Hz.

NOTE B: The equipment shall meet specified performance for the following:

a. Transmitter frequency accuracy of better than 0.002% and power output of 20 watts minimum except the test shall be performed at channels 990, 1100, 1210, 1340, 1470, and 1600. Tests at line frequencies of 47.5 Hz and 420 Hz and high and low line voltage shall be performed at any one channel

b. Receiver noise figure shall not exceed 10 dB with the test performed at channel 1400.

c. Overall frequency response in accordance with Table III of Paragraph 3.5.3.2, except frequencies below 10 kHz be deleted from frequency response Measurement.

d. Transmitter power to the antenna of 13 watts minimum with the tests performed at channels 990, 1360, and 1600.

e. Rise time of the PCM pulse from 10% to 90% amplitude shall be between 10 ns and 50 ns. Decay time of the PCM pulse from 90% to 10% amplitude shall be between 20 ns and100 ns. PCM pulse width at 50% amplitude shall be between 90 ns and 150 ns. The error rate shall be less than 5 errors/second at a received carrier level of -92 dBm.

f. Paragraph 4.10 of this specification.

3.7 Electromagnetic interference. The radio set shall comply with the electromagnetic interference requirements of Appendix B to this specification. For definition of terms, refer to ANSI C63.14-1994.

3.7.1 Bonds and grounds. Bonding for current path return, RF potentials and shock hazard shall be so installed that expansion, contraction or movement incident to normal service use will not break or loosen the connection. Surface preparation for bonds and grounds shall be accomplished by removing all anodic film, grease, paint, lacquer or other high resistance properties from the immediate area of contact. Bonding impedance shall not exceed 2.5

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milliohms. Case mounted connectors shall have an approved RF conductive finish. Bonding impedance of the connector shell to the case shall not exceed 2.5 milliohms. Shielded cables shall be terminated with connectors having an approved RF conductive finish and EMI backshell for 360° peripheral bonding of the shield.

3.8 Reliability.

3.8.1 Numerical reliability requirement. The radio set shall have a mean-time-between-failure (MTBF) of 1200 hours when operated in accordance with Appendix A to this specification as follows:

a. Temperature: 40° ±5°C

b. Temperature cycling: none.

c. Vibration: 2.2G ±10% peak acceleration value at any non-resonant frequency between 20 Hz and 60 Hz, measured at the mounting points of the equipment. The duration of vibration shall be for two minutes of every operating hour during one eight-hour period each week.

d. Equipment on-off cycling: Equipment shall be operated continuously for periods of 144 hours and shall be switched off for periods of eight hours.

e. Input voltage: nominal specified voltage ± 5%.

f. Input voltage cycling: None.

3.9 Safety engineering.

3.9.1 Personnel hazards. Personnel hazards shall be minimized through compliance of configuration changes and parts selection. Recommend use of requirement 1 of MIL-STD-454. Compliance with this requirement will be verified via a visual inspection.

3.9.2 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes, surge arrestors and lenses).

3.10 Marking.

3.10.1 General. Marking shall conform to the applicable drawing. Refer to requirement 67 of MIL-HDBK-454 for guidance.

3.10.2 Visibility. Wherever practicable, parts shall be so mounted that their identification marking will be readily visible with minimum disassembly of the equipment.

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3.10.3 Chassis marking. All chassis identification, parts identification, warning notes, etc., shall be applied by the silk screen process or processes giving the same degree of legibility, placement accuracy and permanence. The marking shall be in accordance with the drawings.

3.10.4 Serial numbers. The following components shall have serial numbers and name plates:

Transmitter, Radio T-983( )P/GRC-103(V)

Amplifier-Frequency Multiplier AM-4322( )/GRC-103(V)

Receiver, Radio R-1329( )P/GRC-103(V)

Amplifier-Converter AM-4318( )/GRC-103(V)

Receiver-Transmitter, Order Wire RT-773( )/GRC-103(V)

Antenna AS-1854 ( )/GRC-103(V)

Mast AB-952( )/GRC-103(V)

3.10.4.1 Composition of serial numbers. The serial numbers used the radio set shall have ten (10) digits. The first five digits shall be the Federal identification number of the prime manufacturer. The sixth digit shall be a code letter to indicate the production contract with said manufacturer. A for first production, B for second production, etc. The final four digits shall be progressively from 0000 to 9999 as required. For substantially continuous production on more than one contract, consecutive numbering without change of letter designator may be used upon application to and approval by the Contracting Officer. Records will be maintained and furnished the Government of the first and last serial number of each unit, assembly, subassembly and module furnished on the contract.

3.11 Finishes. Radio sets and components shall be given a protective finish. This includes finish of hardware type items, such as handles, hinges, screws, etc., and necessary touch-up after mounting. Unless otherwise specified, the final paint film on surfaces exposed to view during operation, shall be a smooth green, in color. When some other color is referred to in the equipment specification, but the details are not specified, the contractor shall apply to the contracting officer for information regarding the required finish. For interference reduction purposes, interface surfaces of structures and bonding connections between cases and mounting bases shall have a finish for electrical contact.

3.12 Power requirements. The radio set shall operate from a 115V rms, 47.5 Hz to 420 Hz, single phase source, having a voltage variation of plus or minus 10%. The total power drain for a transmitter and receiver shall not exceed 400 watts at normal operating conditions.

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3.13 Parts interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically interchangeable without modification or adjustment except operator adjustment of such items or of the radio set.

3.14 Workmanship. All electronic parts, components, assemblies or sub-assemblies shall be free of smudges, excess solder, metal chips or the existence of any foreign material on any surface. Bearing assemblies shall be free from rust, dirt or tool marks. Wires or integrated circuitry shall be protected from contact with rough or irregular surfaces and be shielded from shorting.

3.15 Operation prior to shipment. Each complete radio set shall meet a final operational test in accordance with paragraph 4 and approved test procedure after all conformance tests and inspections have been successfully completed to determine that the radio set is completely operational prior to packaging (See 6.2).

3.16 Technical literature, tools and running spare parts. Technical literature, tools, and running spare parts shall be furnished when so specified in the contract or purchase order. Running spare parts shall be identical to corresponding parts installed in the radio set (see 6.2).

4. VERIFICATION

4.1. Resposibility for inspection. Unless otherwise specified in the contract or puchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any order facilities suitable for the performance of the inspection requirements specified herein, unless is approved by the government. The government reserves the right to perform any of the inspections set forth in this specification whwere such inspections are deemed necessary to ensure suppliers and serviced conform to prescribed requirements.

4.2. Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3); for inspections covered by subsidiary documents (see 4.3).

b. Conformance inspection (see 4.5).

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4.3 First article inspection. First article inspection shall be performed on the first seven (7) units of radio sets (see 3.1). This inspection shall include the visual inspection of 4.3.1 and the tests of 4.3.2.

4.3.1 Visual inspection. Each sample radio set shall be visually examined in accordance with inspections covered by subsidiary documents (para. 4.4) and Table IX.

TABLE IX. Visual inspection.

Defect	Requirement paragraph	Test paragraph
Any part or component missing or damaged	3.4	4.3.1
Systems safety	3.9	4.3.1
Improper marking	3.10	4.3.1
Finishes not as specified	3.11	4.3.1
Parts not interchangeable	3.13	4.3.1
Workmanship of poor quality	3.14	4.3.1

4.3.2 Tests. From those first article samples passing 4.3.1, the tests of Table X shall be performed. Failure of any sample to pass all the tests of Table X shall constitute failure of the first article. The inspections shall be performed in the following order; visual inspection, preconditioning, group A and B for all units, and designated units for specified subgroup within Groups C and D.

TABLE X. First article inspection.

Inspection	Inspection paragraph
All items covered in Table IX visual inspection	4.3.1
Group A inspection	4.5.1
Group B inspection	4.5.2
Group C inspection	4.5.3
Group D inspection	4.5.4
Preconditioning	4.6

4.4 Inspections covered by subsidiary documents. The following items shall be inspected as part of the inspections required by this performance specification and the inspection requirements specified in the contract or purchase order:

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Item	Where Required
Construction	3.4
Finish	3.11
Marking	3.10
Parts interchangeability	3.13
Workmanship	3.14

4.5 Conformance inspection. The contractor shall perform the inspections specified in 4.4, 4.5.1 thru 4.5.5, and 4.6. This does not relieve the contractor of the responsibility for performing any other inspections deemed necessary to contract the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit (and each subassembly and assembly identified in Table VIII) on contract or purchase order shall be inspected for conformance to the inspections.

a. Prior to their assembly into the end equipment, the subassemblies and assemblies identified in Table XI shall be subjected to a 100 percent visual/mechanical and electrical inspection by the contractor for conformance to the inspection requirements provided in Table XI.

b. Subassemblies and assemblies from the discrete lots which have passed the above quality assurance verification inspection shall be assembled into the end equipment which shall then be subjected to sampling inspection in accordance with Table XI. These subassemblies and assemblies shall then be subjected by the contractor to the preconditioning requirements of 4.6 and a 100 percent visual/mechanical and electrical inspection in accordance with Table XI in that order. Lots in which samples exhibit any failures shall be screened for that failure prior to the units within the lot being subjected to Group B inspection.

c. Group A electrical inspection shall be performed in the order shown in Table XI with the exception that the operational test (4.10) shall be last (see 6.2L).

NOTE: Subassemblies and assemblies are not required to undergo further visual/mechanical and electrical inspection as part of the contractor's 100 percent group A inspection of the end equipment.

TABLE XI. Group A inspection.

Inspection	Required paragraph	Test paragraph
(a) Visual and mechanical Receiver, radio (1RE1) R-1329(p)/GRC-103(V) Synthesizer (1RE1A2) Panel, monitor (1RE1A1)	3.4, 3.11 3.10, 3.13, 3.14	4.20

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Power supply (1RE1PS1) Restorer, pulse form (1RE1A3) Limiter-discriminator (1RE1A4) Amplifier-frequency multiplier (1RE1A5) Amplifier, video (1RE1AR1) Amplifier, intermediate freq. (1RE1AR2) Case, receiver (1A2) CY-4638/GRC-103(V) Distribution box (1A2A1)		
	3.10, 3.11	4.20
Amplifier-converter (34) AM-4318( )/GRC-103(V) Duplexer (34A1) Converter, frequency, electronic (34A2) Amplifier, radio frequency (34A7) Control-indicator (34A4) Control-monitor, signal level (34A5)	3.13, 3.4, 3.14	
Transmitter, radio (5TR1) T-983 (P) ( )/GRC-103(V) Synthesizer (5TR1A2) Panel, monitor (5TR1A1) Power supply (5TR1PS1) Control, alarm (5TR1A3) Amplifier-Frequency Multiplier (5TR1A5) Amplifier-Monitor (5TR1A5) Case, transmitter (5A2) CY-4637( )/GRC-103(V) Distribution box (5A2A1)	3.4, 3.10 3.11, 3.13, 3.14	

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TABLE XI. Group A inspection. - Continued.

Inspection	Required paragraph	Test paragraph
Amplifier-frequency multiplier (38) AM-4322/GRC-103(V) Amplifier, radio frequency (38AR1) Frequency multiplier assy (38A1) Control-indicator (38A2)	3.4, 3.10 3.11, 3.13, 3.14	4.20
Receiver-transmitter, Order wire (9) RT-773( )/GRC-103(V) Panel, distribution (9A1) Panel, monitor (9A2) Converter, telephone signal (9A3) Amplifier assembly (9A4) Case, R/Trans, order wire (9MP1)	3.4, 3.10, 3.11 3.13, 3.14	4.20
Antenna (22) AS-1854( )/GRC-103(V) Reflector, antenna (22A1) Antenna element (22A2)	3.4, 3.10, 311 3.13, 3.14	4.20
Mast (12) Mast AB-952( )/GRC-103(V) Elevator, antenna Mast section Accessory kit, mast Bag, mast accessories Guy Ring, guy attachment Spike common Adapter, antenna Anchor, guy Hammer Strap wrench	3.14, 3.10, 3.11, 3.13 3.14	4.20
Cable assemblies Cable assy, R.F. CG-3443/U Cable assy, R.F. CG-3444/U Cable assy, S.P. Elect. CX-10763( )/GRC-103(V)	3.14, 3.10 3.11, 3.13 3.14	4.20

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TABLE XI. Group A inspection. - Continued

Inspection	Required paragraph	Test paragraph
(b) Electrical (Subassemblies)	3.5.7	4.19.1
Receiver, Radio R-1329( )(P)/GRC-103(V)		
Dimmer switch		
Wide band noise		
Frequency deviation		
FDM output		
OW output		
PCM regeneration		
PCM squelch		
PCM mute		
Receiver squelch		
Receiver signal monitor		
FDM return loss		
Video return loss		
Frequency generation		
Power output		
Video frequency response		
Amplifier-converter AM-4318/GRC-103(V)	3.4, 3.5.7	4.19.2
AGC		
Noise figure		
Input VSWR		
Transmitter, radio T-983( )(P)(P)/GRC-103(V)	3.5.7, 4.20	4.19.3
Voltage stability		
Video input and deviation metering		
Power output		
Overheat alarm		
Low power		
Output frequency		
Amplitude-frequency response		
Video input match		
Amplifier frequency multiplier AM-4322/GRC-103(V)	3.5.7	4.19.4
Initial power output		
Voltage check		
Power output and tracking		
Synthesizer (1RE1A2)		
Frequency Check		
Synch Alarm		

TABLE XI. Group A inspection. - Continued

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Inspection	Required paragraph	Test paragraph
Deviation Power output Panel, monitor (1RE1A1) Continuity Check Power supply (1RE1PS1) Initial calibration Meter Check Regulation Load & Regulation Unregulated supply Ripple measurement (unreg.) Ripple measurement (reg.) High freq. instability Restorer, pulse form (1RE1A3) Timing pulse PCM regenerator OW noise test OW distortion Limiter-discriminator (1RE1A4) Demod. linearity Deviation sensitivity Frequency response Amplifier-frequency multiplier 1(RE1A5) Harmonic level Amplifier, video (1RE1AR1) Gain 24 CH PCM Low signal alarm Frequency response Distortion Continuity check Amplifier, intermediate frequency (1RE1AR2) Gain (without AGC) Frequency response AGC limited output level Case, receiver (1A2) CY-4638( )/GRC103(V) Distribution box (1A2) Continuity check	3.10 3.5.7 3.5.7 3.5.7 3.5.7 3.5.7	4.19.8 4.19.9 4.19.6 4.19.7 4.19.10 4.19.7 4.19.12 4.19.14

TABLE XI. Group A inspection. - Continued

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Inspection	Required paragraph	Test paragraph
Amplifier-converter (34) AM-4318( )/GRC-103(V) Tuner, radio frequency (34A1) Duplexer (34A1A1) Adjustments of lead lengths Receiver channel calibration Transmitter channel calibration VSWR bandwidth Receiver Path Insertion Loss Receiver Path Ripple Transmitter Path Insertion Loss Transmitter Path Ripple Local Oscillator Signal Tracking Converter, Frequency Electronic (34A2) Frequency Tracking VSWR Output Power Mixer Stage, Frequency (34A7) Gain and Bandwidth VSWR Noise Figure Control-Indicator (34A4) Coding Check Control Monitor, Signal Level (34A5) VSWR Insertion Loss Isolation Measurement Transmitter, Radio (5TR1) Panel, Monitor (5TR1A1) Continuity Check Synthesizer (5TR1A2) Frequency Check Synch. Alarm Deviation Power Output Control, Alarm (5TR1A3) Low Power Alarm and Backlash Adjustment	3.5.7 3.5.7 3.5.7 3.5.7 3.5.7 3.5.7 3.5.7	4.19.14 4.19.15 4.19.16 4.19.17 4.19.18 4.19.19 4.19.20 4.19.5 4.19.21

TABLE XI. Group A inspection. - Continued

Inspection	Required paragraph	Test paragraph

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:54Z
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Amplifier-Frequency Multiplier (5TR1A4) Frequency Response Output Level Power Output (with ALC) Harmonic Suppression Input VSWR	3.5.7	4.19.22
Amplifier-Monitor (5TR1A5) Operational Check Metering Check Frequency Response	3.5.7	4.19.23
Power Supply (5TR1PS1) Insulation and Resistance Preliminary Check Meter Reading Input Rectification High Tension Rise Time and Frequency Measurement at 5kHz Converter Rise Time and Frequency Measurement at 400 Hz Converter 26V dc Unreg. Supply Filament Supply 28V dc/12 Vdc Reg. Supply High Frequency Instability Constant Current High Tension Metering Circuit Calibration 5 kHz Transistor Saturation Blower Voltage and Alarm Supply Hum Measurement (120 Hz) Spike Measurement (5kHz) Input AC Current Short Circuit Protection Check Starting Circuit Check	3.5.7	4.19.24
Case, Transmitter (5A2) CY-4637( )/GRC-103(V) Distribution Box (5A2A1) Amplifier-Frequency Multiplier (38) AM-4322( )/GRC-103(V)	3.5.7	4.19.25

TABLE XI. Group A inspection. - Continued

Inspection	Required paragraph	Test paragraph
Amplifier, Radio Frequency (38AR1)	3.5.7	4.19.26

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:54Z
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Drive Power Calibration		
Output Amplifier-Power Output		
Output Amplifier-Input Matching		
Driver Amplifier-Power Output		
Driver Amplifier-Input Matching		
Complete Amplifier-Tuning and Power Output		
Frequency Multiplier Assembly (38A1)	3.5.7	4.19.27
RF Alignment		
Level Control		
Input VSWR		
Control-Indicator (38A2)	3.5.7	4.19.28
Coding Check		
Dummy Load (38AR1AT1)	3.5.7	4.19.33
VSWR		
Receiver-Transmitter Order Wire (9)		
RT-773( )/GRC-103(V)		
Panel, Distribution (9A1)	3.5.7	4.19.29
Point-to-point Resistance		
Panel, Monitor (9A2)	3.5.7	4.19.30
Point-to-point Resistance		
Converter, Telephone Signal (9A3)	3.5.7	4.19.31
Oscillator Frequency		
Signal Receiver Selectivity		
Signal Receiver Sensitivity		
Amplifier Assembly (9A4)	3.5.7	4.19.32
Thru Paths		
Power Gain		
Frequency Response		
Harmonic Distortion		
Crosstalk		
Insert Paths		
Power Gain		
Frequency Response		
Harmonic Distortion		
Signal Compression		
Drop Paths		
Power Gain		
Frequency Response		
Harmonic Distortion		
Side Tone Level		

TABLE XI. Group A inspection. - Continued

Inspection	Required paragraph	Test paragraph
(c) Electrical (Main Assemblies)		
Receiver, Radio R-1329(P)( )/		

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GRC-103(V) with Amplifier-Converter AM-4318( )/GRC-103(V)		
Demodulation	3.5.2.3	4.15
Noise Figure	3.5.2.2	4.15
IF Bandwidth	3.5.2.5	4.19.1
Received Signal Monitoring	3.5.2.4	4.15
Receiver Protection	3.5.2.6	4.18
(c) Electrical (Main Assemblies)		
Receiver, Radio R-1329(P)( )/ GRC-103(V) with Amplifier-Converter AM-4318( )/GRC-103(V) (Cont.)		
Frequency Response Shaping	3.5.2.1	4.15
Voltage Standing Wave Ratio (Duplexes)	3.5.3.11	4.19.15
Transmitter, Radio T-983(P)( )/ GRC-103(V) with Amplifier-Frequency Multiplier AM-4322( )/GRC-103(V)		
Frequency Response	3.5.1.5	4.16
Power Output	3.5.1.1	4.16
Power Dissipation	3.5.1.3	4.16
Setability	3.5.1.4	4.16
Low Power Alarm	3.5.1.2	4.16
Receiver-Transmitter, Order Wire (RT-773( )/GRC-103(V)		
Tone Generator	3.5.4.1.1	4.17
Deviation	3.5.4.3	4.19.3
Side Tone	3.5.4.11	4.19.32
Peak Limiting	3.5.4.5	4.19.32
Antenna (AS-1854( )/GRC-103(V))		
Symmetry	3.5.5.1	4.13
Squint	3.5.5.2	4.13
Minor Lobes	3.5.5.3	4.13
Gain	3.5.5.4	4.13
VSWR	3.5.5.5	4.13
Impedance	3.5.5.6	4.13
Electrical (System)		
Radio Set AN/GRC-103(V)3		
Operational	3.15	4.10

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order specified in the contract or purchase order.

4.5.2 Group B inspection. Group B inspection shall be performed on lots that have passed Group A inspection. This inspection shall consist of inspections listed in Table XII. Each lot shall be subjected to sampling inspection in accordance with Table XII. Lots in which samples

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exhibit any failures shall be screened for that failure prior to the units within the lot being subjected to Group C inspection.

4.5.2.1 Order of inspection within Group B. Group B inspection shall be performed in the order listed in Table XII.

TABLE XII. Group B inspection.

Inspection	Required paragraph	Test paragraph
Electrical (Main Assemblies)		
Receiver, Radio R-1329( )(P)/GRC-103(V) with Amplifier-Converter AM-4319( )/GRC-103(V)		
Output Impedance response	3.5.3.2	4.19.1
Squelch and alarm	3.5.3.7.3	4.19.1
Transmitter, Radio T-983( )(P)/GRC-103(V) with Amplifier-Frequency Multiplier AM-4323( )/GRC-(V)		
Input Impedance	3.4.3.1	4.19.3
PCM Modulation	3.4.3.4	4.19.3
Electrical (System)		
Radio Set AN/GRC-103(V)3		
Frequency Response	3.5.3.3	4.18
PCM Signal-to-Noise	3.5.3.5	4.18
PCM Distortion	3.5.3.6	4.18
Pulse Regeneration	3.5.3.7.1	4.18
Timing Signal	3.5.3.7.2	4.18
Order Wire Distortion	3.5.4.8	4.18
Order Wire Signal-to-Noise	3.5.4.4	4.18
Power Requirement	3.5.3	4.11, 4.18
Order Wire Frequency Response	3.5.4.6	4.18
Order Wire Net Loss	3.5.4.10	4.22
Duplexer	3.5.3.9	4.19.15
Duplexer Discrimination	3.5.3.10	4.19.15
Interchangeability		
Parts	3.13	4.21.1, 4.21.2

4.5.3 Group C inspection. Group C inspection shall be performed on units which have passed Group A and Group B inspections. The inspections shall consist of the inspections specified in Table XIII below. Sampling shall be selected in accordance with 4.5.3.1.

TABLE XIII. Group C inspection.

Inspection	Required paragraph	Test paragraph

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:54Z
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Radio Set Components		
Sub-Group I		
Antenna	3.5.5	4.13
Symmetry	3.5.5.1	4.13
Squint	3.5.5.2	4.13
Minor lobes	3.5.5.3	4.13
Gain	3.5.5.4	4.13
Mast		
Erection	3.5.6.1	4.14
Operation	3.5.2.2	4.14
Transmitter modulation	3.5.1.7	4.19.3
Duplexer insertion loss	3.5.3.9	4.19.1.5
Systems level stability	3.5.3.12	4.9.1
Shock, bench-handling	3.6.4	4.7.3
Bounce test - vehicular	3.6.4	4.7.1
Sub-Group II		
Low temperature	3.6.1.1	4.7.4.1
High temperature	3.6.1.2	4.7.4.2
Altitude	3.6.3	4.7.6
Humidity	3.6.2	4.7.5
Sub-Group III		
Electromagnetic interference	3.7	4.23
Shock test, ballistic	3.6.4	4.7.2

4.5.3.1 Sampling for Group C inspection. Sampling shall be performed in accordance with the contract or purchase order.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in the order specified in 4.5.3.1.

4.5.3.3 Non-compliance. The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of the failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Defense Contract Management Officer (DCMO). After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspection which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

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4.5.3.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be acceptable on the contract provided all damage is repaired and the same units are resubjected and pass group A inspection only.

4.5.4 Group D inspection. This inspection shall consist of the test specified in Table XIV and shall be performed on units from lots which have been subjected to and met groups A and B inspection.

TABLE XIV. Group D inspection.

Inspection	Required paragraph	Test paragraph
Reliability	3.8	4.12

4.5.4.1 Sampling for Group D inspection. Samples shall be selected at random in accordance with 4.12 and the Appendix. For each production lot, successful completion of the reliability testing must be determined before delivery.

4.5.4.2 Group D failures. Actions required relative to Group D failure shall be as specified in the contract or purchase order.

4.5.4.3 Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and passed group D inspection may be accepted on the contract or purchase order provided all visible and known damage is repaired and the sample units are resubjected to and pass Group A inspection only.

4.6 Preconditioning.

4.6.1 Bounce/Vibration preconditioning. The unit shall be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tested, shafts in place, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the units shall not be repaired, aligned, cleaned or otherwise changed prior to subjection to acceptance inspection. All equipment delivered on the contract or order shall be subjected to the requirements of this paragraph. If an equipment fails any test because of the bounce preconditioning, the defect applies to that test.

4.6.2 Electrical preconditioning.

4.6.2.1 Sequence of testing. Before subjecting equipment to the following procedure, the contractor shall have available objective evidence of compliance to Group A requirements.

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4.6.2.2 Test conditions. Every equipment shall be subjected to a 48 hour burn-in period The last 24 hours of burn-in shall be failure free. The burn-in shall be conducted as if it were the beginning of a Production Reliability Test of the Appendix to this specification. The failures shall be recorded and analyzed, but shall not be considered relevant for computation of equipment MTBF.

4.6.2.3 Compliance. Units passing burn-in debugging shall be submitted to Groups A, B, and C inspection in that order without benefit or repair, cleaning realignment, or other changes, unless otherwise specified herein. If, after successful completion of burn-in debugging, the equipment fails either the Group A or B tests, the equipment will be reworked and resubmitted to burn-in debugging.

4.7 Service conditions. (Radio only): For antenna test procedures see SM-A-696455, and for the Mast, see SM-A-696070.

4.7.1 Bounce test: Class L units. The equipment shall be tested in accordance with 6.3, or equivalent validation method. Prior to testing, the equipment shall meet full specification performance for the measurements specified in Table VII. During Part I, all connections (such as cable assemblies) shall be in place. When practicable, package tester accessories shall be selected from those listed on Drawing SC-A-46439. The adapter plate shall be free of cracks and broken welds and all unused holes, except the 2-inch breather holes, shall be plugged. No part of the mounted surface shall be less than 7 inches from the edge of the adapter plate. Additional barriers used to safeguard personnel shall not restrain the fore-and-aft motion of the adapter plate against the backstop.

4.7.2 Shock test: ballistic. The test shall be test in accordance with 6.3, or equivalent validation method. The equipment, including shock mounts (if any), shall be secured in its normal operating position to the steel test plate by means of the same fasteners used for vehicular installation of the equipment. The test shall consist of a total of 9 blows: one each 1-foot blow, 3-foot blow, and 5-foot elbow on the back, side and top of the test plate. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible. Prior to testing, the equipment shall meet full specification performance for the measurements specified in Table VII.

4.7.3 Shock test: bench-handling. The equipment shall be tested in accordance with 6.3, or equivalent validation method. Prior to an after testing, the equipment shall meet full Specification performance for the measurements specified in Table VII.

4.7.4 Temperature tests.

4.7.4.1 Low temperature. The equipment shall be tested in accordance with the table in 6.3, or equivalent validation method. The storage temperature shall be -65°F and the operating temperature shall be -35°F. Prior to, during and after testing, the equipment shall meet full specification performance for the measurements specified in 4.9.1.

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4.7.4.2 High temperature. The equipment shall be tested in accordance with the table in 6.3, or equivalent validation method. The operating temperature shall be 125°F. Prior to during and after testing, the equipment shall meet full specification performance for the measurements specified in 4.9.1.

4.7.5 Humidity (moisture-resistance) test. The equipment shall be tested in accordance with the table in 6.3, or equivalent validation method. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 4.9.1. Measurements shall be made during the periods shown on the figure for the cycle. Do not move equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If any equipment fails to meet the specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment still fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test.

4.7.6 Altitude (elevation) test. The equipment shall be tested in accordance with 6.3, or equivalent validation method, except the chamber internal pressure in Step 2 shall be 282.0 mm of HG (11.1 in. of Hg. or 25,000 ft above sea level). During and after testing, the equipment shall meet full specification performance, and humidity unless otherwise specified.

4.8 Standard inspection conditions. Unless otherwise specified, all tests shall be performed at 115 V + 5% from a single-phase source operating between 47.5 and 63.0 Hz. Tests shall be conducted at room ambient temperature, pressure, and humidity unless otherwise specified.

4.9 Test specification (service conditions). The equipment shall be tested in accordance with the following test specifications and meet the requirements of this specification. The tests shall be performed during periods as specified in 6.3.

4.9.1 Temperature and humidity. Item 1.3 and 1.4 of SM-A-764246 (except the test) shall be performed at channels 990, 1100, 1210, 1340, 1470 and 1600 instead of those specified. Tests at line frequencies of 47.5 Hz and 420 Hz and high and low line voltage shall be performed at any one channel. Item 4.1 of SM-A-764245 with the exception that tests shall be performed at channels 990, 1360, and 1600 instead of those specified. Item 1.2 of SM-A-764247; except that test shall be performed at channel 1400 instead of that specified. Item 4.2 of SM-A-764245, except frequencies below 10 kHz shall be deleted from frequency response Measurement. Item 4.5 of SM-A-764245. Item 4.6.8 of SM-A-764245. Item 4.7 of SM-A-764245.

4.9.2 Altitude. Item 1.4 of SM-A-764246, except that the test shall be performed at channels 1000 and 1400 instead of those specified. Item 1.3 of SM-A-764246, except the test shall be performed at channel 1400 instead of the channel specified. Item 1.7 of SM-A-764246, except that the test shall be performed at channel 1600 instead of the channels specified. Item 1.4 of SM-A-764247, except that the test shall be performed at channel 1300 instead of the channels specified. Item 4.5 of SM-A-764245.

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4.10 Operational test service conditions. With the equipment set up in a system, an operational test shall be performed. The test shall consist of voice communications, order wire ringing through the systems plus a visual check of the metering facilities for reach specified in 4.1.6 and 4.2.1.2.6 of SM-A-764245.

4.11 AC Power line, voltage and frequency tests. The following tests shall be performed at room ambient and at channels 40, 180 and 400. The tests shall be performed at 105 V 47 Hz, 115 V 47 Hz, 125 V 47 Hz, 105 V 420 Hz, 115 V 420 Hz and 125 V 420 Hz.

Inspection	Requirement paragraph
1.3 and 1.4 of SM-A-764246	3.5.1
1.4 of SM-A-764247	3.5.2.2
4.2.5 of SM-A-698826	3.5.3.7
4.2.2 of SM-A-764245	3.5.3.4
4.10 of this specification	3.5.4.1

4.12 Reliability testing.

4.12.1 Reliability qualification test. The contractor shall perform a reliability qualification test in accordance with Test Plan IV of MIL-STD-781 and the Appendix to this specification. The minimum test time per equipment should follow the procedure of paragraph 4.2.2.4 of MIL-HDBK-781.

4.12.2 Production reliability sampling. After successfully passing the Reliability Qualification Test, production reliability sampling shall begin. The production Sampling Tests shall commence with the first lot available after the models used for first article testing. The samples should be tested in accordance with Test Plan IVA of MIL-HDBK-781 and the Appendix to this Specification. The specified (0) MTBF shall be 1200 hours. The minimum test time per equipment should be in accordance with paragraph 4.2.2.4 of MIL-HDBK-781.

4.12.3 Testing. All testing shall be conducted under the environment of Test Level B of MIL-HDBK-781, or equivalent validation method. Equipment on-off cycling shall be on for 144 hours, off 8 hours. Vibration shall be for two minutes of every operating hour during one eight-hour period each week.

4.12.4 Failure definition. A failure is defined as the inability of the equipment to perform its required function. This includes any event which causes departure from performance as required by this Specification or the contractors approved test procedures. Paragraph 5.5 through 5.10 inclusive of MIL-HDBK-781 shall apply. The types of failures and required actions shall be in accordance with paragraph 5.5 of MIL-HDBK-781. Determination of compliance may be in accordance with paragraph 5.4.8 of MIL-HDBK-781, or equivalent validation method.

4.12.5 Rework and retest provisions. Once a reject decision has been reached, no rerun can be made of that test on the same or new sample from the lot unless positive action consisting

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of design change, modification, or rework actually relating to observed failures has been applied to the lot. Any subsequent rerun must start at zero hours operating time.

4.13 Antenna test procedure. That antenna shall be tested in accordance with SM-A-696455, or equivalent validation method, and meet the requirements of 3.5.5.

4.14 Mast test procedures. The mast shall be tested in accordance with SM-A-696070, or equivalent validation method, and meet the requirements of 3.5.6.

4.15 Radio set AN/GRC-103(V) 3 receiver section test procedure. The receiver section shall be tested in accordance with paragraphs 1.3, 1.4, 1.5, and 1.6 of SM-A764247, or equivalent validation method, and meet the requirement of 3.5.2.1; 3.5.2.2.; 3.5.2.3, 3.5.3.8, and 3.5.4.2; and 3.5.2.4 respectively.

4.16 Radio set AN/GRC 103( ) (V)3 transmitter section test procedure. The transmitter section shall be tested in accordance with paragraphs 1.3, 1.4, 1.5, 1.6, and 1.7 of SM-A-764246, or equivalent validation method, and meet the requirements of 3.5.1.4, 3.5.1.1, 3.5.1.2, 3.5.1.3, and 3.5.1.5 respectively.

4.17 Receiver-transmitter, order wire RT-773( ) GRC-103(V) test. The receiver-transmitter, order wire shall be tested in accordance with 4.2.5 of SM-A-698807, or equivalent validation method, and meet the requirement of 3.5.4.1.1

4.18 Radio set AN/GRC-103( )(V)3 test procedure. The radio set shall be tested in accordance with paragraphs 4.2.3, 4.3, 4.4, 4.5, 4.6.7, 4.6.8, 4.7, and 4.8 of SM-A-764245, or equivalent validation method, and meet the requirements of 3.5.3.3; 3.5.3.5; 3.5.3.6; 3.5.3.7.1, and 3.5.3.7.2; 3.5.4.8; 3.5.4.4; 3.5.2.6; and 3.12 respectively.

4.19 Component test. The assemblies, subassemblies and modules of Radio Set, AN/GRC-103(V) shall be tested in accordance with the following drawings in order to demonstrate compliance with 3.13 of this specification. Specific designations of test frequencies specified herein and/or contractor's prepared test procedures shall be construed as examples to show minimum number of frequencies selected to determine compliance and shall not be construed as restricting testing at other than the listed frequencies. Each requirement shall apply equally to all frequencies within the frequency range. Testing shall be performed on designated frequencies plus six (6) randomly selected frequencies spaced across the band as selected by the contracting officer's technical representative (see 6.1.1).

4.19.1 Receiver, Radio R-1329( )(P)/GRC-103(V). The Receiver, Radio, R-1329( )(P)/GRC-103(V) shall be tested in accordance with SM-A-698826, or equivalent validation method, and meet the requirements thereof. Meeting paragraph 4.2.15 of this drawing demonstrates compliance with 3.4.2.5. Tests for 3.5.3.2 and 3.5.3.7.3 are found in paragraphs 4.2.12 and 4.2.8 of SM-A-883588 respectively (see 6.4).

4.19.2 Amplifier-converter AM-4318/GRC-103(V). The Amplifier-Converter,

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AM-4318( )/GRC-103(V) shall be tested in accordance with SM-A-698656, or equivalent validation method, and meet the requirements thereof(see 6.4)..

4.19.3 Transmitter, Radio T-983( )(P)/GRC-103(V). The Transmitter, Radio, T-983( )(P)/GRC-103(V) shall be tested in accordance with SM-A-883590, or equivalent validation method, and meet the requirements thereof. Tests for the requirements of paragraphs 3.5.4.3, 3.5.3.1, 3.5.3.4 and 3.5.1.7 are found in paragraphs 4.3.5, 4.9, 4.3.1 through 4.3.3 and 4.3.4 of SM-A-883590. (see 6.4).

4.19.4 Amplifier-frequency multiplier AM 4322/GRC-103(V). The Amplifier Frequency Multiplier AM-4322( )/GRC-103(V) shall be tested in accordance with SM-A-698660, or equivalent validation method, and meet the requirements thereof. (see 6.4).

4.19.5 Synthesizer (1RE1A2/5TR1A2). The synthesizer shall be tested in accordance with SM-A-865041, or equivalent validation method, and meet the requirements thereof. (see 6.4).

4.19.6 Restorer, Pulse Form (1RE1A3). The Restore, Pulse Form shall be tested in accordance with SM-A-698241, or equivalent validation method, and meet the requirements thereof. (see 6.4).

4.19.7 Limiter-Discriminator (1RE1A4). The Limiter-Discriminator shall be tested in accordance with SM-A-698031, or equivalent validation method, and meet the requirements thereof. (see 6.4).

4.19.8 Panel, Monitor (1RE1A1). The Panel, Monitor shall be tested for continuity in accordance with SM-A-698317, or equivalent validation method, and meet the requirements thereof. (see 6.4).

4.19.9 Power Supply (1RE1PS1). The Power Supply shall be tested in accordance with SM-A-698372, or equivalent validation method, and meet the requirements thereof. (see 6.4).

4.19.10 Amplifier-Frequency Multiplier (1RE1A5). The Amplifier-Frequency Multiplier shall be tested in accordance with SM-A-698610, or equivalent validation method, and meet the requirements thereof. (see 6.4).

4.19.11 Amplifier, Video (1RE1AR1). The Amplifier, Video shall be tested in accordance with SM-A-698035, or equivalent validation method, and meet the requirements thereof. (see 6.4).

4.19.12 Amplifier, Intermediate (1RE1AR2). The Amplifier, Intermediate shall be tested in accordance with SM-A-698002, or equivalent validation method, and meet the requirements thereof. (see 6.4).

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4.19.13 Distribution Box (1A2A1). The Distribution Box shall be tested for continuity in accordance with SM-A-698685, or equivalent validation method, and meet the requirements thereof. (see 6.4).

4.19.14 Tuner, Radio Frequency (34A1). The Tuner shall be tested in accordance with SM-A-698656, or equivalent validation method and meet the requirements thereof. (see 6.4).

4.19.15 Duplexer (34A1A1). The Duplexer shall be tested in accordance with SM-A-696387, or equivalent validation method, and meet the requirements thereof. Tests for the requirements of paragraphs 3.5.3.11, 3.5.3.9 and 3.5.3.10 are found in paragraphs 5.2; 5.3, 5.4; 5.5, 5.6 and 5.7 of SM-A-696387, respectively. (see 6.4).

4.19.16 Converter, Frequency, Electronic (34A2). The Converter, Frequency, Electronic shall be tested in accordance with SM-A-698652, or equivalent validation method. The appropriate subassemblies shall be tested in accordance with SM-A-696396, SM-A-696397, SM-A-698781, or equivalent validation method, and meet the respective requirements thereof. (see 6.4).

4.19.17 Mixer Stage, Frequency (34A7). The Mixer Stage, Frequency shall be tested in accordance with SM-A-69673-2, or equivalent validation method, and meet the requirements thereof. (see 6.4).

4.19.18 Control Indicator Coding Check (34A4). The Control Indicator shall be tested in accordance with SM-A-696394, or equivalent validation method, and meet the requirements thereof. (see 6.4).

4.19.19 Control Monitor, Signal Level (34A5). The Control Monitor shall be tested in accordance with SM-A-698643, or equivalent validation method, and meet the requirements thereof. (see 6.4).

4.19.20 Panel, Monitor (5TR1A1). The Panel Monitor shall be tested in accordance with SM-A-698790, or equivalent validation method, and meet the requirements thereof. (see 6.4).

4.19.21 Control, Alarm (5TR1A3). The Control, Alarm shall be tested in accordance with SM-A-698776, or equivalent validation method, and meet the requirements thereof. (see 6.4).

4.19.22 Amplifier-Frequency Multiplier (5TR1A4). The Amplifier-Frequency Multiplier shall be tested in accordance with SM-A-696443, or equivalent validation method, and meet the requirements thereof. (see 6.4).

4.19.23 Amplifier-Monitor (5TR1A5). The Amplifier-Monitor shall be tested in accordance with SM-A-698664, or equivalent validation method, and meet the requirements thereof. (see 6.4).

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4.19.24 Power Supply (5TR1PS1). The Power Supply shall be tested in accordance with SM-A-698363, or equivalent validation method, and meet the requirements thereof. (see 6.4).

4.19.25 Distribution Box (5A2A1). The Distribution Box shall be tested in accordance with SM-A-698789, or equivalent validation method, and meet the requirements thereof. (see 6.4).

4.19.26 Amplifier, Radio (38AR1). The Amplifier, Radio shall be tested in accordance with SM-A-698659, or equivalent validation method, and meet the requirements thereof. (see 6.4).

4.19.27 Frequency Multiplier Assembly (38A1). The Frequency Multiplier Assembly shall be tested in accordance with SM-A-698657, or equivalent validation method, and meet the requirements thereof. (see 6.4).

4.19.28 Control-Indicator (38A2). The Control-Indicator shall be tested in accordance with SM-A-696394, or equivalent validation method, and meet the requirements thereof. (see 6.4).

4.19.29 Panel, Distribution (9A1). The Panel, Distribution shall be tested in accordance with SM-A-698759, or equivalent validation method, and meet the requirements thereof. (see 6.4).

4.19.30 Panel, Monitor (9A2). The Panel, Monitor shall be tested in accordance with SM-A-698695, or equivalent validation method, and meet the requirements thereof. (see 6.4).

4.19.31 Converter, Telephone Signal (9A3). The Converter, Telephone Signal shall be tested in accordance with SM-A-698805, or equivalent validation method, and meet the requirements thereof. (see 6.4).

4.19.32 Amplifier Assembly (9A4). The Amplifier Assembly shall be tested in accordance with SM-A-698806, or equivalent validation method, and meet the requirements thereof. Recommended tests for the requirements of 3.4.4.11 and 3.4.4.5 are found in paragraphs 4.1.2, 4.1.5, 4.2.2, 4.3.2, 4.3.5; and 4.2.5 of SM-A-698806. (see 6.4).

4.19.33 Dummy Load (38AR1AT1). The Dummy Load shall be tested in accordance SM-A-698658, or equivalent validation method, and meet the requirements thereof. (see 6.4).

4.20 Visual and mechanical inspection. Parts and equipment shall be examined for the defects listed herein. In addition, the equipment shall be visually examined for compliance with the requirements of system safety (3.9).

4.21 Interchangeability inspection.

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4.21.1 Mechanical interchangeability. Mechanical interchangeability shall be demonstrated by disassembling all normally removable modules from two units that have been subjected to and meet all other Group B tests except electrical interchangeability, interchanging approximately one-half of the modules (while assuring that one pair of each type module is interchanged) and reassembling. The main frame assemblies (fixed and tuning heads) shall be interchanged between bases. The units shall meet the requirements of 3.13. The above method is applicable for first article tests. In production, mechanical interchangeability shall be demonstrated by gauging the applicable parts, assemblies and units with the gages listed on GL-SM-B-323451 and subordinate lists.

4.21.2 Electrical interchangeability. Electrical interchangeability shall be demonstrated by disassembling all normal removable modules from Group B tests, interchanging approximately one-half of the modules (while assuring that one pair of each type module is interchanged) and reassembling. The main frame assemblies (Fixed and tuning heads) shall be inter-changed between cases. The units shall then be operated and measurements made of the transmitter power output and the receiver noise quieting at two frequencies in the band. The units shall meet the requirements of 3.5.1.1 and 3.5.2.2.

4.22 Order wire operational tests. Two terminals shall be set up to operate in a full duplex PCM system connected through sufficient attenuation. Frequency response and net loss measurements shall be made with the equipment meeting the requirements of 3.5.4.6 and 3.5.4.10.

4.23 Electromagnetic interference tests. The AN/GRC-103(V)3 shall be tested for compliance to the electromagnetic interference requirements of 3.4.1.6, 4.3.2.7 and 3.12. Broadband emission tests shall be performed with the AN/GRC-103( )( )3 operating at 850 MHz. All tests except CE06 shall be performed with the equipment operating in a full duplex mode.

4.23.1 Bonding and grounding tests. Prior to the start of EMI tests, compliance with the requirement of 3.7.1 shall be verified. Bonding data shall be included in the EMI test report.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activity within the Military Department of Defense Agency, or within the Military Department’s System Command. Packaging data retrieval is available from the managing Military Department’ or Defense Agency’s automated packaging files, CD-ROM products or by contacting the responsible packaging activity.

6. NOTES

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(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. Radio Set, AN/GRC(V)3, is used to provide simultaneous, two-way (duplex) transmissions. It is a complete, compact, rugged radio relay set that may be used either as a terminal or frequency modulation patch repeater for intermediate or forward area systems. It provides radio-frequency (RF) channel assignments from 695 MHz to 1000 MHz.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Type required.

c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2 and 2.3).

d. First article inspection and number of samples required (see 3.1).

e. Levels of preservation and packaging (see section 5).

f. First article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

g. First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of times to be tested as specified in 4.3. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

h. Group C inspection. Approval to ship may be withheld at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action (see 4.5.3.3).

i. Initial shipments. Initial shipment of end items will be contingent only upon successful completion of all group C test (see 4.5.3).

j. Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a

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letter identifying the particular design for example: AN/GRC-103(V)3. The contractor shall apply for nomenclature in accordance with the applicable clause in the contract.

k. Color. The color chip furnished by the procuring agency will be the color chip Number 24098 and may be obtained upon request to the contracting officer.

l. Location of operational tests. It is desirable that operational system tests be performed at a location that will minimize handing after these tests are completed (see 4.10).

m. Other. Technical literature, tools and running spare parts.

6.3 Environmental inspection procedures. The test methods and procedures of MIL-STD-810 listed below are recommended, but not mandatory for determining compliance with the environmental requirements of this specification. The contractor may use any test method that he can verify as providing comparable stress levels on the materiel to be tested.

Inspection	Spec Paragraph	MIL-STD-810 Method	MIL-STD-810 Procedure	Remarks
Low temperature	3.6.1.1	502	I	Storage temp. shall be -65°F and the operating temp. shall be -35°F
High temperature	3.6.1.2	501	II	Storage and operating temp. shall be +125°F.
Humidity	3.6.2	507	II	
Altitude	3.6.3	500	I	The chamber internal pressure (step 2) shall be 282.0mm of HG (11-1 in. of Hg. or 25,000 ft.)
Vibration/Bounce (Class L Units)	3.6.4.1	514	IX	Prior to and after test, the radio set shall meet full performance for the measurements in Table VIII.
High intensity shock	3.6.4.2	516.2	II	(See above)
Shock bench handling	3.6.4.3	516	V	(See above)

6.4 Electrical inspection procedures. The test methods and procedures of the drawings referenced below are recommended, but not mandatory for determining compliance with the requirements of Paragraphs 3.5.7.1 through 3.5.7.33 of this specification. The contractor may use any test method that he can verify as providing comparable test results on the materiel to be tested.

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Component/Subassembly	Requirement Paragraph	Referenced Test Procedure
Receiver, Radio R-1329 ()(P)/GRC-103(V).	3.5.7.1	SM-A-698826
Amplifier-converter AM-4318/GRC-103(V).	3.5.7.2	SM-A-698656
Transmitter, Radio T-983( )(P)/GRC-103(V)	3.5.7.3	SM-A-883590
Amplifier-frequency Multiplier AM-4322/GRC-103(V)	3.5.7.4	SM-A-698660
Synthesizer (1RE1A2/5TR1A2)	3.5.7.5	SM-A-865041
Restorer, Pulse Form (1RE1A3)	3.5.7.6	SM-A-698241
Limiter-Discriminator (1RE1A4)	3.5.7.7	SM-A-698031
Panel Monitor (1RE1A1)	3.5.7.8	SM-A-698317
Power Supply (1RE1PS1).	3.5.7.9	SM-A-698372
Amplifier-Frequency Multiplier (1RE1A5)	3.5.7.10	SM-A-698610
Amplifier, Video (1RE1AR1)	3.5.7.11	SM-A-698035
Amplifier, Intermediate (1RE1AR2).	3.5.7.12	SM-A-698002
Distribution Box (1A2A1).	3.5.7.13	SM-A-698685
Tuner, Radio Frequency (34A1).	3.5.7.14	SM-A-69856
Duplexer (34A1A1).	3.5.7.15	SM-A-696387
Converter, Frequency, Electronic (34A2).	3.5.7.16	SM-A-698652, SM-A-696396, SM-A-696397, and SM-A-698781
Mixer Stage, Frequency (34A7).	3.5.7.17	SM-A-696763-2
Control Indicator Coding Check (34A4).	3.5.7.18	SM-A-696394
Control Monitor, Signal Level (34A5).	3.5.7.19	SM-A-69843
Panel, Monitor (5TR1A1).	3.5.7.20	SM-A-698790
Control, Alarm (5TR1A3).	3.5.7.21	SM-A-698776
Amplifier-Frequency Multiplier (5TR1A4).	3.5.7.22	SM-A-696443
Amplifier-Monitor (5TR1A5).	3.5.7.23	SM-A-698664
Power Supply (5TR1PS1).	3.5.7.24	SM-A-698363
Distribution Box (5A2A1).	3.5.7.25	SM-A-698789
Amplifier, Radio (38AR1).	3.5.7.26	SM-A-698659
Frequency Multiplier Assembly (38A1)	3.5.7.27	SM-A-698657
Control-Indicator (38A2).	3.5.7.28	SM-A-696394

Component/Subassembly	Requirement Paragraph	Referenced Test Procedure
Panel, Distribution (9A1).	3.5.7.29	SM-A-698759
Panel, Monitor (9A2).	3.5.7.30	SM-A-698695
Converter, Telephone Signal (9A3).	3.5.7.31	SM-A-698805
Amplifier Assembly (9A4).	3.5.7.32	SM-A-698806

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Dummy Load (38AR1AT1).	3.5.7.33	SM-A-698658

6.5 Subject term (key word) listing.
Equipment, relay, radio
Repeater, patch, modulation frequency
Set, relay, radio
Terminal

6.6 Changes from previous issue. Marginal notation are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CUSTODIAN:
Army-CR

PREPARING ACTIVITY:
Army-CR

(Project 5820-0150)

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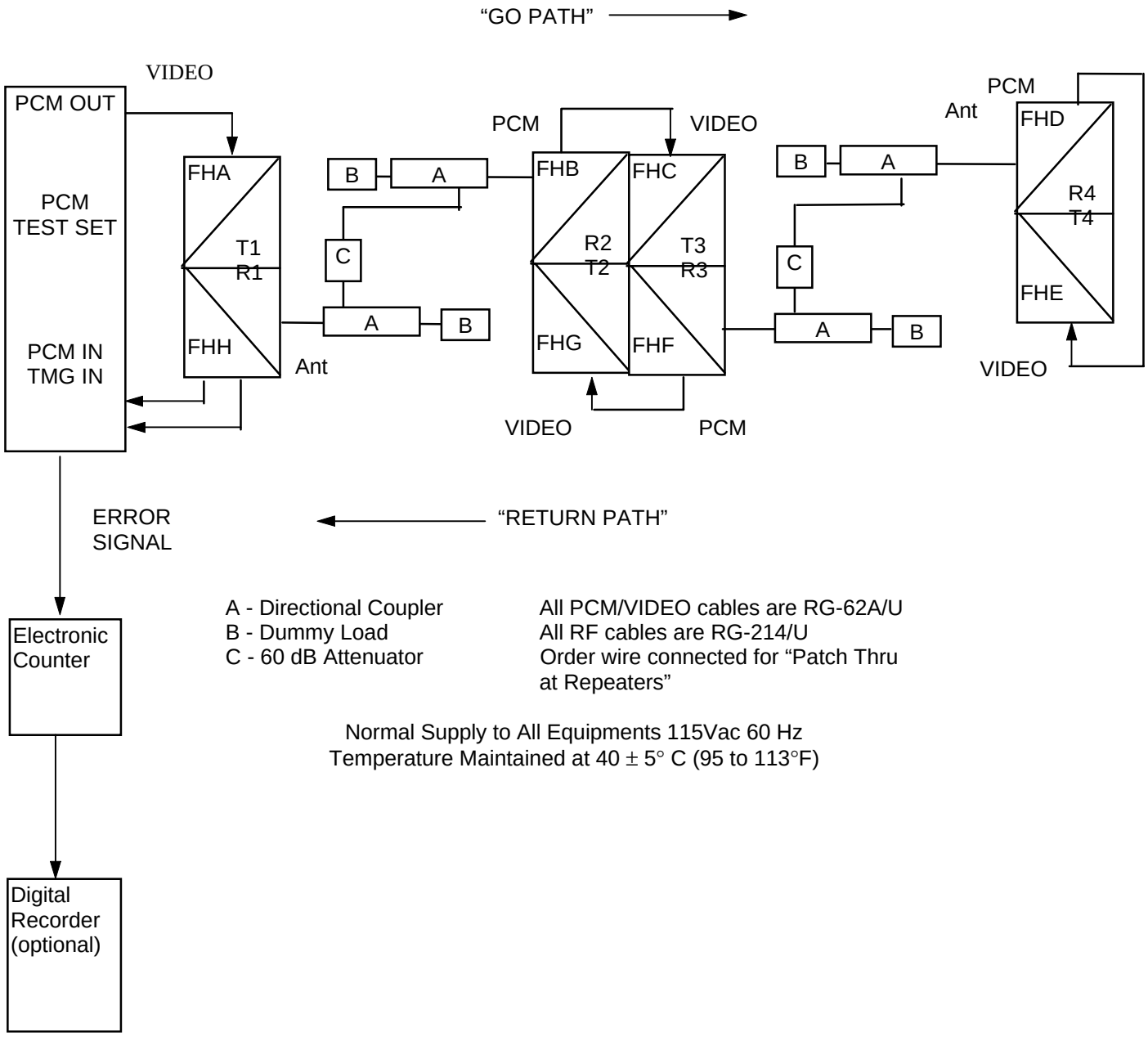


FIGURE 1. Reliability test arrangement AN/GRC-103(V).

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APPENDIX A

RELIABILITY VERIFICATION PROGRAM FOR RADIO SET AN/GRC-103(V)3

A.1 SCOPE

A.1.1 Scope. This appendix outlines and specifies the requirements for the reliability verification program applicable to Radio Set AN/GRC-103(V)3. It is a mandatory part of the specification. The information contained herein is intended for compliance.

A.1.2 The reliability test program for the Radio Set, AN/GRC-103(V)3 shall be completed as per the following test plans:

a. Reliability Qualification Test, in accordance with MIL-HDBK-781, Test Plan IV, on four (4) systems to be selected from the First Article equipment.

b. Production Reliability Sampling Test, in accordance with MIL-HDBK-781, Test Plan IVA, or equivalent validation method, on four (4) production systems selected from every other month's production lot.

A.1.3 The Reliability Demonstration shall be completed after reaching the acceptance criteria of plan IV or IVA (as applicable) of MIL-STD-781, or equivalent validation method. The MTBF (0) for the above plans shall be 1200 hours of the system.

A.2. APPLICABLE DOCUMENTS

Not applicable to this appendix.

A.3. TEST CONDITIONS

A.3.1 The four equipment shall be set up in a two-way, two-hop system as shown in Figure 1 of Specification MIL-PRF-49185B(CR) using suitable attenuators to absorb the output power from each transmitter and to provide an appropriate level of signal to each following receiver. The four equipment shall carry a PCM signal and any deterioration in performance of any piece of equipment in the system shall produce error signals at the receiving end. The faulty equipment will then be located by checking individual meter readings.

A.3.2 The test shall be conducted in a suitable test chamber maintained at 40° ±5°C (95% to 113°F).

A.3.3 The radio equipment shall be accessible to an operator for switching and returning as required, and for interchanging of the R.F. Heads as called for in the test procedure.

A.3.4 The radio equipment shall operate continuously for periods of 144 hours and shall be switched off for periods of eight hours.

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During these “off periods” the R.F. Heads shall be interchanged as detailed in sub-section 40. A time recording meter shall be used to show total running time for the equipment.

A.3.5 For two minutes of every operating hour during the eight hour period each week, each equipment shall be vibrated at 2.2 g + 10% peak acceleration value at any non-resonant frequency between 20 Hz and 60 Hz, measured at the mounting points of the equipment.

A.3.6 The equipment undergoing reliability testing shall be subjected to a burn-in period of at least 48 hours prior to the start of reliability testing. This period of equipment operation shall be performed in accordance with applicable portions of the equipment specification.

A.4. VERIFICATION PROCEDURE

A.4.1 With the four radio sets connected as shown in Figure 1 of Specification MIL-PRF-49185B(CR), set them to the first group of channels shown in Table XIII.

A.4.2 With PCM test set on, adjust the transmitter No. A input control to provide +180 kHz peak deviation as indicated by the panel meter reading (in the green band when switched to 12 CH. PCM).

A.4.3 Check the meter reading of the following receiver No. B when switched to 12 Channel. PCM. It should show a reading in the green band.

A.4.4 Adjust the input control of transmitter No. C for +180 kHz peak deviation as indicated on the panel meter and check the following receiver No. D.

A.4.5 Repeat for each equipment group in the “forward” path and then back on the “return” path. Under normal conditions there shall be no appreciable error signal output from the PCM test set and the electronic counter shall only give random indications. In the event of a fault developing anywhere in the system, error signals shall appear at the output and shall give visual indications on the counter. In order to provide continuous unattended monitoring, it is desirable that a digital recorder (summation type) be connected to the output of the counter.

A.4.6 Every four hours through the test period, the operating channels shall be changed in accordance with the sequence in Table XIII. Channel changes shall continue until an Accept/Reject decision is reached (see MIL-HDBK-781, Tables IV and IVa).

A.4.7 Every six days the R.F. Tuning Heads shall be interchanged in accordance with the schedule in Table XIV. This will be carried out during the off periods defined in subsection 30. Tuning Heads shall be changed in the following Manner:

a. The fixed heads shall be designated alphabetically and shall remain in fixed positions with the case. The R.F. heads shall be designated numerically and shall be moved as required throughout the demonstration.

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b. The identifying numbers are prefixed by the letter T or R to indicate whether they transmitting or receiving heads.

c. If a fourth (4) change is required, Table XIV will revert back to the start position and the sequence shall begin again: This shall continue as long as changes are required.

A.4.8 The meter readings of all equipment shall be logged together with information on R.F. Heads and channels used, after each frequency change. Order wire signaling shall be checked daily between all equipment on test.

A.5. FAILURE ACTION.

A.5.1 Only failures occurring within the equipment under test directly as a result of normal operation shall be considered as relevant. Failures in the following categories are classed as non-relevant:

a. Failures in any of the test equipment used.

b. Failures due to accidents or improper test procedures.

c. Failures occurring during the burn-in period.

d. Failures which can be eliminated by mutually agreed corrections.

e. Failures caused by other failures.

A.5.2 When a suspected relevant failure occurs in any equipment, the test shall be stopped and the failed equipment removed, repaired, and reinstalled at which time testing may continue.

A.5.3 The failed equipment shall be appropriately tagged to show the nature of the failure and the accumulated hours of operation. A tag shall be retained on the equipment and a notation of time of return to test added. Full details of the failure shall be recorded as detailed in section 60.0 and the failure shall be recorded.

A.6. TEST RECORDS (TO BE MAINTAINED BY THE CONTRACTOR).

A.6.1 A complete operating log shall be kept during the test to record all pertinent information such as operating hours, off periods, vibration periods, channel changes, RF head changes, meter readings, and times at which failures occur or when units are restored to test.

A.6.2 In addition to this operating log, an equipment failure record shall be kept for every equipment under test.

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APPENDIX A

A.6.3 To assist in the demonstration of the MTBF for the equipment, a failure summary record shall be compiled and shall include all failures classed as relevant on all equipment under test.

TABLE XII. List of recommended test equipment.

Description	Manufacturer	Type
Vibration Machine 0-60 Hz	Lab Corporation	Model 30-3000 or equivalent
Accelerometer Random	Columbia Labs	Model 1700 or equivalent
PCM Generator		USACECOM DWG SM-E-698042 or equivalent
Electronic Counter	Hewlett-Packard	5245L or equivalent
Frequency Converter	Hewlett-Packard	5254A/B or equivalent
Directional Coupler	Hewlett-Packard	Model 778D or equivalent
Attenuator 60 dB, 2W	Hewlett-Packard	8491A-660 or equivalent

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APPENDIX A

TABLE XIII. Operating channel settings at 4-hour periods.

Elapsed Hours of Operating Time	RF T1 R2	Tuning T2 R1	Head	Change T3 R4	Pairs T4 R3
0	990	1297		990	1297
4	1451	1144		1451	1144
8	991	1298		991	1298
12	1452	1145		1452	1145
16	992	1299		992	1299
20	1453	1146		1453	1146
24	993	1300		993	1300
28	1454	1147		1454	1147
32	994	1301		994	1301
36	1455	1148		1455	1148
40	995	1302		995	1302
44	1456	1149		1456	1149
48	996	1303		996	1303
52	1457	1150		1457	1150
56	997	1304		997	1304
60	1458	1151		1458	1151
64	998	1305		998	1305
68	1459	1152		1459	1152
72	999	1306		999	1306
76	1460	1153		1460	1153
80	1000	1307		1000	1307
84	1461	1154		1461	1154
88	1001	1308		1001	1308
92	1462	1155		1462	1155
96	1002	1309		1002	1309
100	1463	1156		1463	1156
104	1003	1310		1003	1310
108	1464	1157		1464	1157
112	1004	1311		1004	1311
116	1465	1158		1465	1158
120	1005	1312		1005	1312
124	1466	1159		1466	1159
128	1006	1313		1006	1313
132	1467	1160		1467	1160
136	1007	1314		1007	1314

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TABLE XIII. Operating channel settings at 4-hour periods - Continued.

Elapsed Hours of Operating Time	RF T1 R2	Tuning T2 R1	Head	Change T3 R4	Pairs T4 R3
140	1468	1161		1468	1161
144	1008	1315		1008	1315
148	1469	1162		1469	1162
152	1009	1316		1009	1316
156	1470	1163		1470	1163
160	1010	1317		1010	1317
164	1471	1164		1471	1164
168	1011	1318		1011	1318
172	1472	1165		1472	1165
176	1012	1319		1012	1319
180	1473	1166		1473	1166
184	1013	1320		1013	1320
188	1474	1167		1474	1167
192	1014	1321		1014	1321
196	1475	1168		1475	1168
200	1015	1322		1015	1322
204	1476	1169		1476	1169
208	1016	1323		1016	1323
212	1477	1170		1477	1170
216	1017	1324		1017	1324
220	1478	1171		1478	1171
224	1018	1325		1018	1325
228	1479	1172		1479	1172
232	1019	1326		1019	1326
236	1480	1173		1480	1473
240	1020	1327		1020	1327
244	1481	1174		1481	1174
248	1021	1328		1021	1328
252	1482	1175		1482	1175
256	1022	1329		1022	1329
260	1483	1176		1483	1176
264	1023	1330		1023	1330
268	1484	1177		1484	1177
272	1024	1331		1024	1331
276	1485	1178		1485	1178

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TABLE XIII. Operating channel settings at 4-hour periods - Continued.

Elapsed Hours of Operating Time	RF T1 R2	Tuning T2 R1	Head	Change T3 R4	Pairs T4 R3
280	1025	1332		1025	1332
284	1486	1179		1486	1179
288	1026	1333		1026	1333
292	1487	1180		1487	1180
296	1027	1334		1027	1334
300	1488	1181		1488	1181
304	1028	1335		1028	1335
308	1489	1182		1489	1182
312	1029	1336		1029	1336
316	1490	1183		1490	1183
320	1030	1337		1030	1337
324	1491	1184		1491	1184
328	1031	1338		1031	1338
332	1492	1185		1492	1185
336	1032	1339		1032	1339
340	1493	1186		1493	1186
344	1033	1340		1033	1340
348	1494	1187		1494	1187
352	1034	1341		1034	1341
356	1495	1188		1495	1188
360	1035	1342		1035	1342
364	1496	1189		1496	1189
368	1036	1343		1036	1343
372	1497	1190		1497	1190
376	1037	1344		1037	1344
380	1498	1191		1498	1191
384	1038	1345		1038	1345
388	1499	1192		1499	1192
392	1039	1346		1039	1346
396	1500	1193		1500	1193
400	1040	1347		1040	1347
404	1501	1194		1501	1194
408	1041	1348		1041	1348
412	1502	1195		1502	1195
416	1042	1349		1042	1349

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TABLE XIII. Operating channel settings at 4-hour periods - Continued.

Elapsed Hours of Operating Time	RF T1 R2	Tuning T2 R1	Head	Change T3 R4	Pairs T4 R3
420	1503	1196		1503	1196
424	1043	1350		1043	1350
428	1504	1197		1504	1197
432	1044	1351		1044	1351
436	1505	1198		1505	1198
440	1045	1352		1045	1352
444	1506	1199		1506	1199
448	1046	1353		1046	1353
452	1507	1200		1507	1200
456	1047	1354		1047	1354
460	1508	1201		1508	1201
464	1048	1355		1048	1355
468	1509	1202		1509	1202
472	1049	1356		1049	1356
476	1510	1203		1510	1203
480	1050	1357		1050	1357
484	1511	1204		1511	1204
488	1051	1358		1051	1358
492	1512	1205		1512	1205
496	1052	1359		1052	1359
500	1513	1206		1513	1206
504	1053	1360		1053	1360
508	1514	1207		1514	1207
512	1054	1361		1054	1361
516	1515	1208		1515	1208
520	1055	1362		1055	1362
524	1516	1209		1516	1209
528	1056	1363		1056	1363
532	1517	1210		1517	1210
536	1057	1364		1057	1364
540	1518	1211		1518	1211
544	1058	1365		1058	1365
548	1519	1212		1519	1212
552	1059	1366		1059	1366
556	1520	1213		1520	1213

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TABLE XIII. Operating channel settings at 4-hour periods - Continued.

Elapsed Hours of Operating Time	RF T1 R2	Tuning T2 R1	Head	Change T3 R4	Pairs T4 R3
560	1060	1367		1060	1367
564	1521	1214		1521	1214
568	1061	1368		1061	1368
572	1522	1215		1522	1215
576	1062	1369		1062	1369
580	1523	1216		1523	1216
584	1063	1370		1063	1370
588	1524	1217		1524	1217
592	1064	1371		1064	1371
596	1525	1218		1525	1218
600	1065	1372		1065	1372
604	1526	1219		1526	1219
608	1066	1373		1066	1373
612	1527	1220		1527	1220
616	1067	1374		1067	1374
620	1528	1221		1528	1221
624	1068	1375		1068	1375
628	1529	1222		1529	1222
632	1069	1376		1069	1376
636	1530	1223		1530	1223
640	1070	1377		1070	1377
644	1531	1224		1531	1224
648	1071	1378		1071	1378
652	1532	1225		1532	1225
656	1072	1379		1072	1379
660	1533	1226		1533	1226
664	1073	1380		1073	1380
668	1534	1227		1534	1227
672	1074	1381		1074	1381
676	1535	1228		1535	1228
680	1075	1382		1075	1382
684	1536	1229		1536	1229
688	1076	1383		1076	1383
692	1537	1230		1537	1230
696	1077	1384		1077	1384

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TABLE XIII. Operating channel settings at 4-hour periods - Continued.

Elapsed Hours of Operating Time	RF T1 R2	Tuning T2 R1	Head	Change T3 R4	Pairs T4 R3
700	1538	1231		1538	1231
704	1078	1385		1078	1385
708	1539	1232		1539	1232
712	1079	1386		1079	1386
716	1540	1233		1540	1233
720	1080	1387		1080	1387
724	1541	1234		1541	1234
728	1081	1388		1081	1388
732	1542	1235		1542	1235
736	1082	1389		1082	1389
740	1543	1236		1543	1236
744	1083	1390		1083	1390
748	1544	1237		1544	1237
752	1084	1391		1084	1391
756	1545	1238		1545	1238
760	1085	1392		1085	1392
764	1546	1239		1546	1239
768	1086	1393		1086	1393
772	1547	1240		1547	1240
776	1087	1394		1087	1394
780	1548	1241		1548	1241
784	1088	1395		1088	1395
788	1549	1242		1549	1242
792	1089	1396		1089	1396
796	1550	1243		1550	1243
800	1090	1397		1090	1397
804	1551	1244		1551	1244
808	1091	1398		1091	1398
812	1552	1245		1552	1245
816	1092	1399		1092	1399
820	1553	1246		1553	1246
824	1093	1400		1093	1400
828	1554	1247		1554	1247
832	1094	1401		1094	1401
836	1555	1248		1555	1248

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TABLE XIII. Operating channel settings at 4-hour periods - Continued.

Elapsed Hours of Operating Time	RF T1 R2	Tuning T2 R1	Head	Change T3 R4	Pairs T4 R3
840	1095	1402		1095	1402
844	1556	1249		1556	1249
848	1096	1403		1096	1403
852	1557	1250		1557	1250
856	1097	1404		1097	1404
860	1558	1251		1558	1251
864	1098	1405		1098	1405
868	1559	1252		1559	1252
872	1099	1406		1099	1406
876	1560	1253		1560	1253
880	1100	1407		1100	1407
884	1561	1254		1561	1254
888	1101	1408		1101	1408
892	1562	1255		1562	1255
896	1102	1409		1102	1409
900	1563	1256		1563	1256
904	1103	1410		1103	1410
908	1564	1257		1564	1257
912	1104	1411		1104	1411
916	1565	1258		1565	1258
920	1105	1412		1105	1412
924	1566	1259		1566	1259
928	1106	1413		1106	1413
932	1567	1260		1567	1260
936	1107	1414		1107	1414
940	1568	1261		1568	1261
944	1108	1415		1108	1415
948	1569	1262		1569	1262
952	1109	1416		1109	1416
956	1570	1263		1570	1263
960	1110	1417		1110	1417
964	1571	1264		1571	1264
968	1111	1418		1111	1418
972	1572	1265		1572	1265
976	1112	1419		1112	1419

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APPENDIX A

TABLE XIII. Operating channel settings at 4-hour periods - Continued.

Elapsed Hours of Operating Time	RF T1 R2	Tuning T2 R1	Head	Change T3 R4	Pairs T4 R3
980	1573	1266		1573	1266
984	1113	1420		1113	1420
988	1574	1267		1574	1267
992	1114	1421		1114	1421
996	1575	1268		1575	1268
1000	1115	1422		1115	1422
1004	1576	1269		1576	1269
1008	1116	1423		1116	1423
1012	1577	1270		1577	1270
1016	1117	1424		1117	1424
1020	1578	1271		1578	1271
1024	1118	1425		1118	1425
1028	1579	1272		1579	1272
1032	1119	1426		1119	1426
1036	1580	1273		1580	1273
1040	1120	1427		1120	1427
1044	1581	1274		1581	1274
1048	1121	1428		1121	1428
1052	1582	1275		1582	1275
1056	1122	1429		1122	1429
1060	1583	1276		1583	1276
1064	1123	1430		1123	1430
1068	1584	1277		1584	1277
1072	1124	1431		1124	1431
1076	1585	1278		1585	1278
1080	1125	1432		1125	1432
1084	1586	1279		1586	1279
1088	1126	1433		1126	1433
1092	1587	1280		1587	1280
1096	1127	1434		1127	1434
1100	1588	1281		1588	1281
1104	1128	1435		1128	1435
1108	1589	1282		1589	1282
1112	1129	1436		1129	1436
1116	1590	1283		1590	1283
1120	1130	1437		1130	1437
1124	1591	1284		1591	1284
1128	1131	1438		1131	1438

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APPENDIX A

TABLE XIII. Operating channel settings at 4-hour periods - Continued.

Elapsed Hours of Operating Time	RF T1 R2	Tuning T2 R1	Head	Change T3 R4	Pairs T4 R3
1132	1592	1285		1592	1285
1136	1132	1439		1132	1439
1140	1593	1286		1593	1286
1144	1133	1440		1133	1440
1148	1594	1287		1594	1287
1152	1134	1441		1134	1441
1156	1595	1288		1595	1288
1160	1135	1442		1135	1442
1164	1596	1289		1596	1289
1168	1136	1443		1136	1443
1172	1597	1290		1597	1290
1176	1137	1444		1137	1444
1180	1598	1291		1598	1291
1184	1138	1445		1138	1445
1188	1599	1292		1599	1292
1192	1139	1446		1139	1446
1196	1600	1293		1600	1293
1200	1140	1447		1140	1447
1204	990	1294		990	1294
1208	1141	1448		1141	1448
1212	991	1295		991	1295
1216	1142	1449		1142	1449
1220	992	1296		992	1296
1224	1143	1450		1143	1450

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TABLE XIV. R.F. Tuning heads interchange schedule.

R.F. Head IDENT NBRS.	Fixed Head Identifying Letters				
	START	FIRST CHANGE	SECOND CHANGE	THIRD CHANGE	FOURTH CHANGE
T1	A	G	C	E	A
R2	B	F	D	H	B
T3	C	E	A	G	C
R4	D	H	B	F	D
T4	E	A	G	C	E
R3	F	D	H	B	F
T2	G	C	E	A	G
R1	H	B	F	D	H

CUSTODIAN:
Army - CR

PREPARING ACTIVITY:
Army - CR

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APPENDIX B

ELECTROMAGNETIC INTERFERENCE (EMI) VERIFICATION PROGRAM FOR RADIO SET AN/GRC-103(V)3

B.1. SCOPE

B.1.1 Scope. This appendix outlines and specifies the requirements for the Electromagnetic Interference (EMI) Verification Program applicable to Radio Set, AN/GRC-103(V)3. It is a mandatory part of the specification. The information contained herein is intended for compliance.

B.1.2 The EMI Verification Program for the Radio Set, AN/GRC-103(V)3 shall be completed as per the following test plans:

a. EMI Qualification Test Plan procedures are to be performed on first article equipments as specified in the contract or order.

b. EMI Production Verification Test Plan procedures are to be performed on samples of production equipments as specified in the contract or order.

B.2. APPLICABLE DOCUMENTS

Not applicable to this appendix.

B.3. TEST CONDITIONS

B.3.1 The equipment shall be set up in a shielded enclosure with the test equipment and its related antennas and sensing equipment.

B.3.2 The test shall be conducted at normal room ambient temperature of 20° C to 30°C (68°F to 86°F).

B.3.3 The equipment under test shall be accessible to the operator for switching and retuning as required.

B.4. VERIFICATION PROCEDURE

B.4.1 Conducted emissions, antenna terminal. The equipment is to be tested for conducted emissions on the antenna terminal over the frequency range hereby defined using the measurement setup of Figure B-1. The start frequency shall be as follows:

Operating frequency range	Start frequency of test
300 MHz to 3 GHz	1 MHz

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The end frequency shall be 20 GHz. The conducted emissions at the antenna terminal of the equipment under test shall not exceed the values below.

a. Receivers: 34 dBµV

b. Transmitters (standby mode): 34 dBµV

c. Transmitters (transmit mode): Harmonics, except the second and third, and all other spurious emissions shall be at least 80 dB down from the level at the fundamental. The second and third harmonics shall be suppressed 50 + 10 log p (where p = peak power input in watts at the fundamental) or 80 dB, whichever requires less suppression (CE106).

B.4.2 Conducted susceptibility, power leads, 30 Hz to 50 KHz. The equipment is to be tested for conducted susceptibility on the power leads over the frequency range of 30 Hz to 50 KHz using the measurement setup of Figure B-2. The equipment shall not exhibit any malfunction, degradation of performance, or deviation from specified indications, beyond the tolerances indicated in this specification, when subjected to the test signals shown in Figure B-3 (CS101).

B.4.3 Conducted susceptibility, bulk cable injection, impulse excitation. The equipment is to be tested for conducted susceptibility to using the setup of Figure B-4. The equipment shall not exhibit any malfunction, degradation of performance, or deviation from specified indications, beyond the tolerances indicated in this specification, when subjected to the calibrated test signal shown in Figure B-5 at a repetition rate of 30 Hz for one minute (CS115).

B.4.4 Radiated emissions, electric field, 2 MHz to 18 GHz. The equipment is to be tested for radiated emissions of the electric field over the frequency range of 2 MHz to 18 GHz using the measurement setup of Figure B-6. The emissions shall not exceed the levels shown in Figure B-7. Above 30 MHz, the limits shall be met for both horizontally and vertically polarized fields (RE102).

B.4.5 Radiated susceptibility, electric field, 10 KHz to 40 GHz. The equipment is to be tested for radiated susceptibility to the electric field over the frequency range of 10 KHz to 40 GHz using the measurement setup of Figure B-8. The equipment shall not exhibit any malfunction, degradation of performance, or deviation from specified indications, beyond the tolerances indicated in this specification, when subjected to the electric fields shown below in Table B-I. Up to 30 MHz, the limits shall be met for vertically polarized fields. Above 30 MHz, the limits shall be met for both horizontally and vertically polarized fields. Circular polarized fields are not acceptable (RS103).

Table B-I. Radiated susceptibility limits - electric field

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Frequency range	Limit level (volts/meter)
10 KHz-2 MHz	20
2 MHz - 30 MHz	50
30 MHz - 1 GHz	50
1 GHz - 18 GHz	50
18 GHz - 40 GHz	50

CUSTODIAN:
Army - CR

PREPARING ACTIVITY:
Army - CR

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APPENDIX B

(CE106)

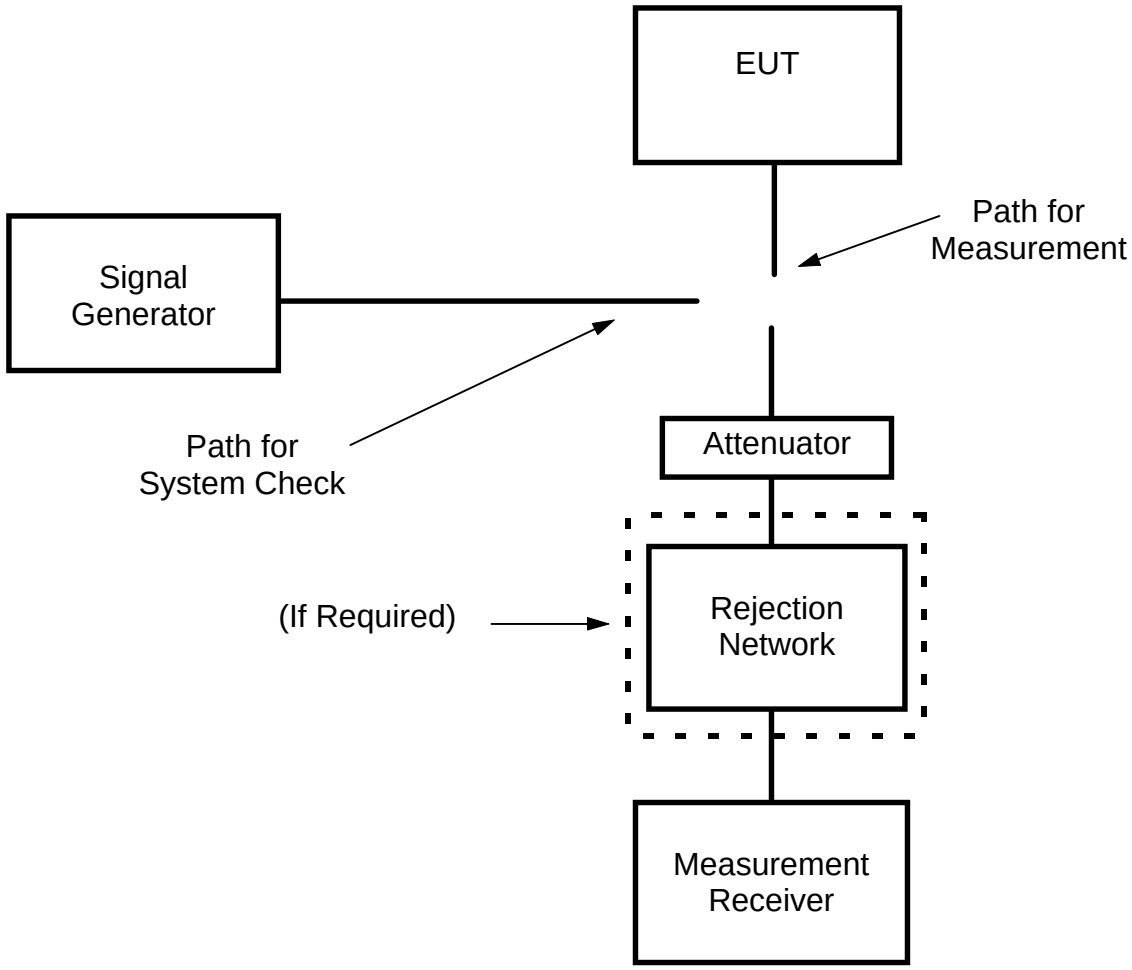


FIGURE B-1. Measurement setup, conducted emissions, antenna terminals.

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(CE106)

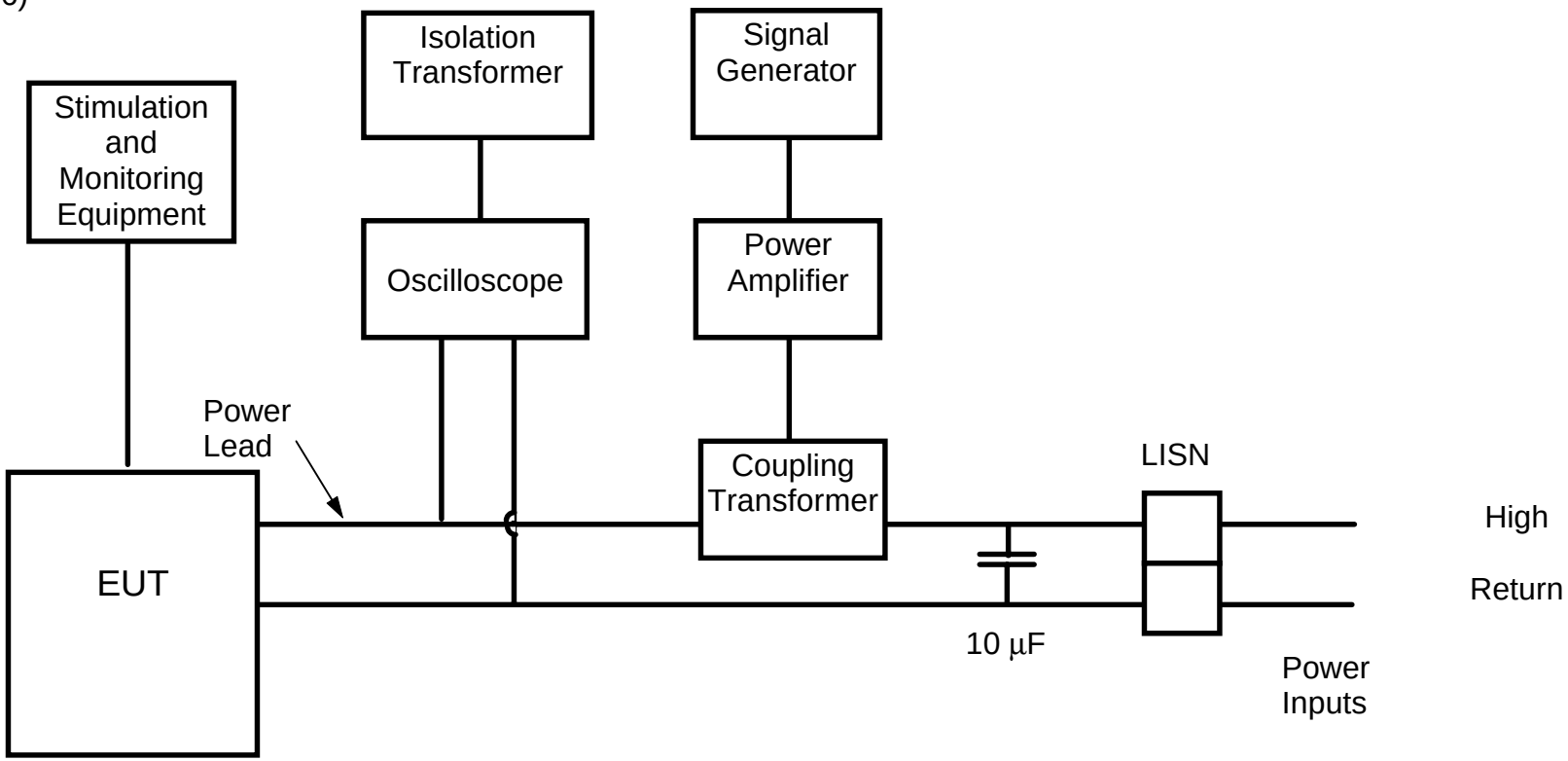
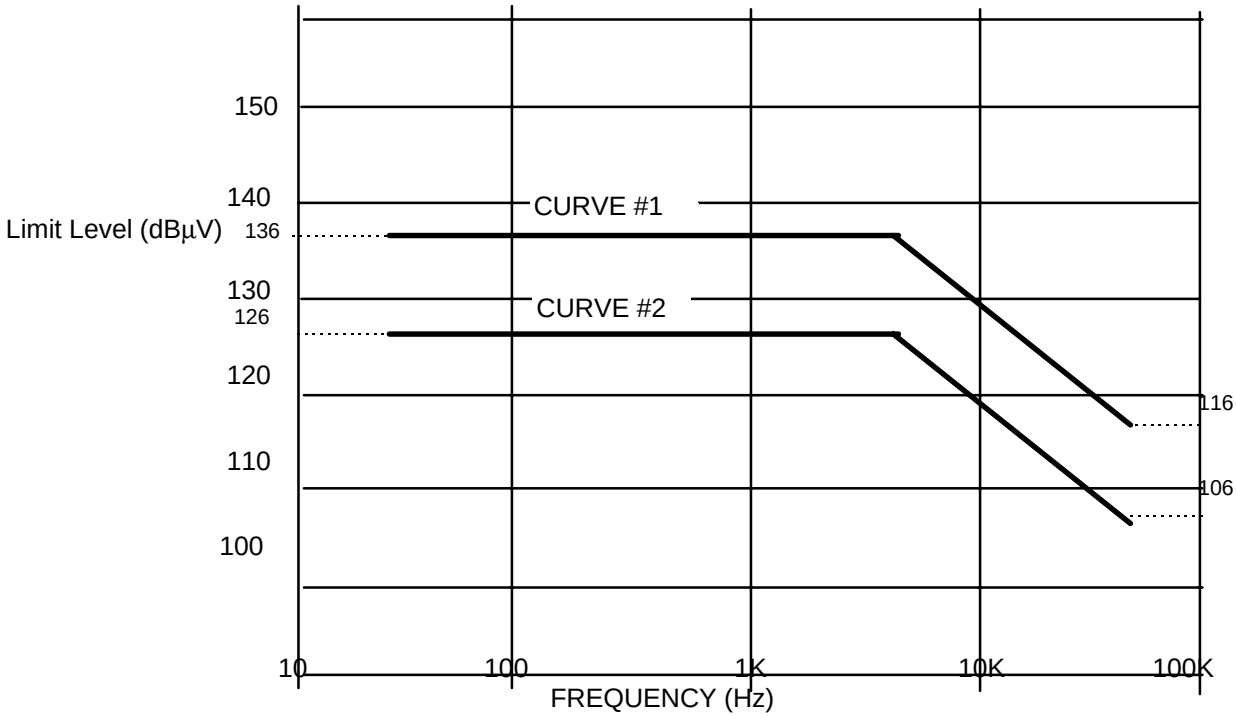


Figure B-2. Signal injection, conducted susceptibility, DC or single phase AC power leads.

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APPENDIX B

(CS101)



NOMINAL EUT SOURCE VOLTAGE	APPLICABLE CURVE
ABOVE 28 VOLTS	#1
28 VOLTS OR BELOW	#2

Figure B-3. Conducted Susceptibility limit, power leads

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APPENDIX B

(CS115)

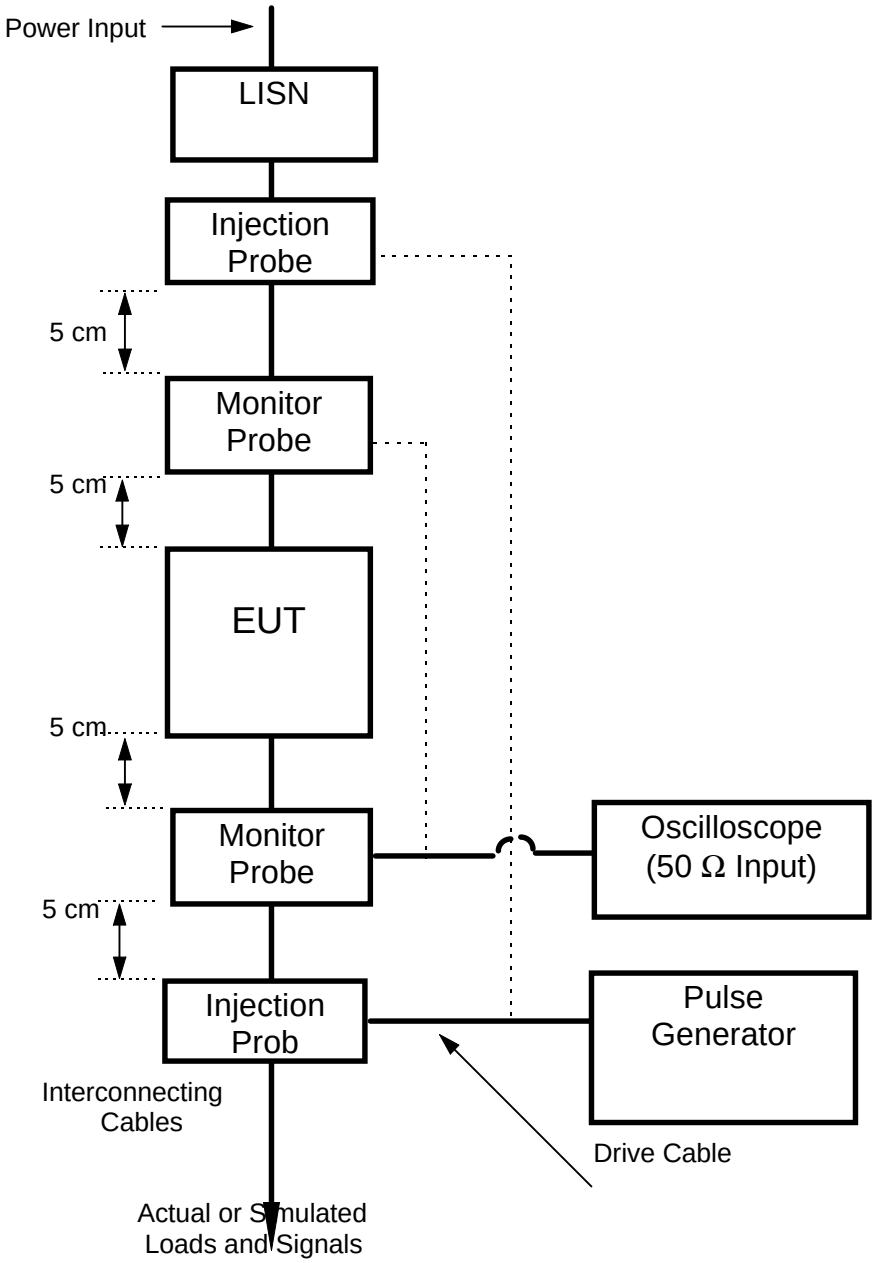


FIGURE B-4. Test setup, conducted susceptibility, bulk cable injection.

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(CS115)

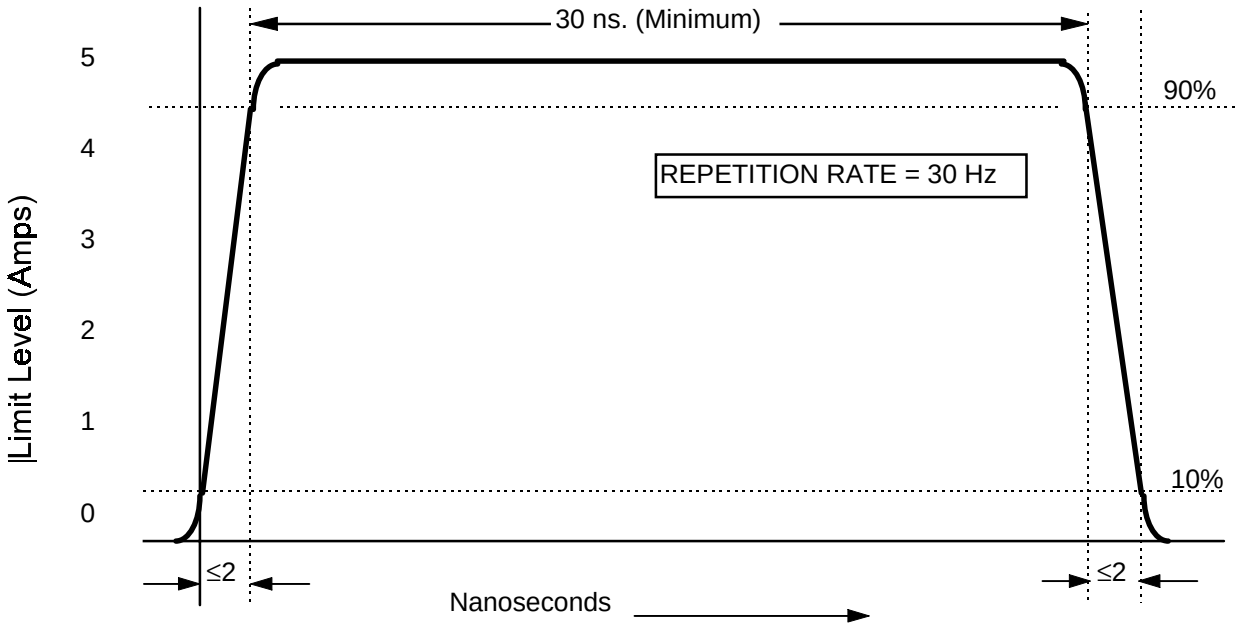


FIGURE B-5. Calibrated signal source characteristics for all applications.

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(RE102)

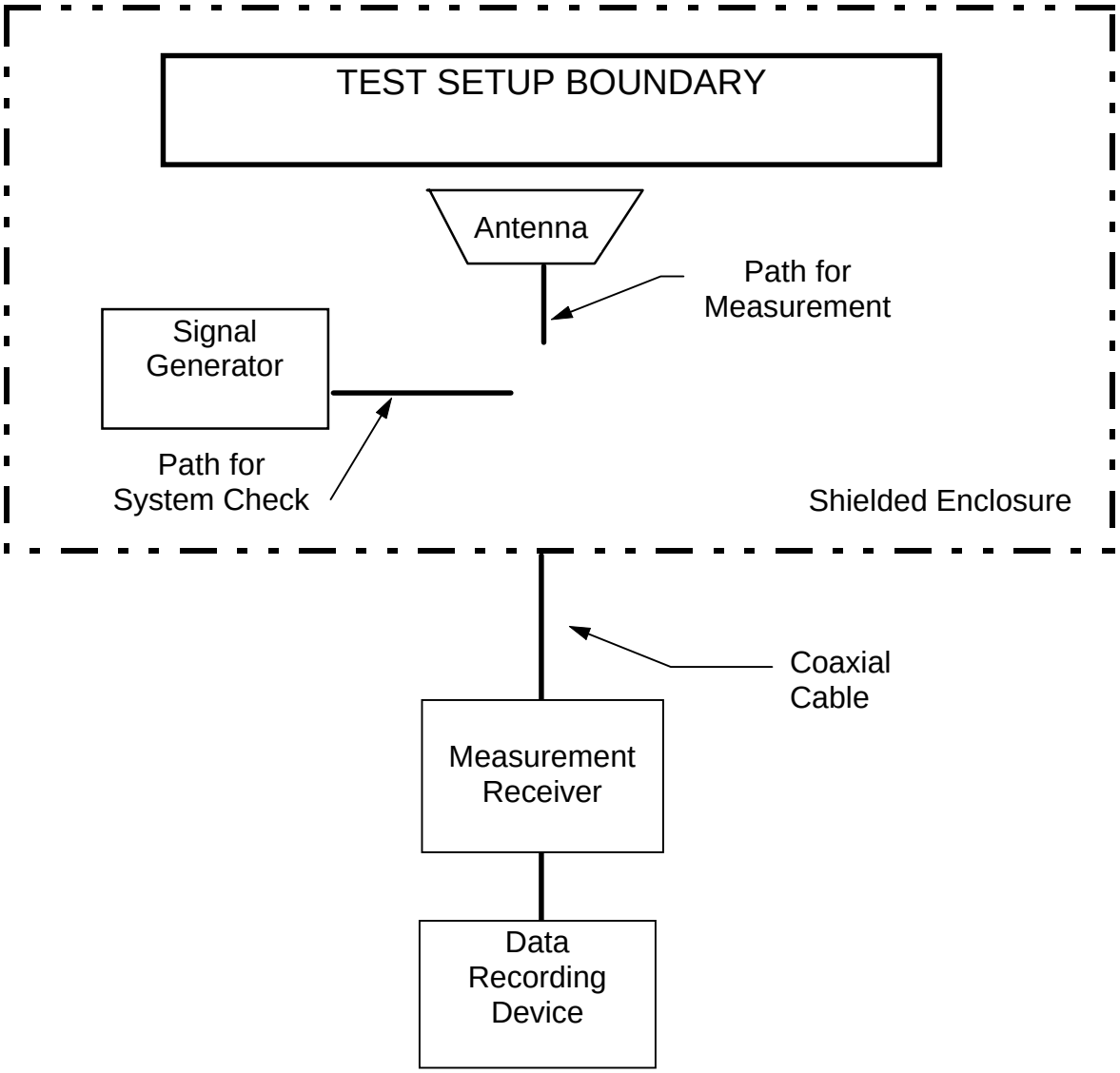


FIGURE B-6. Test setup, radiated emissions, electric field, 2 Mhz to 18 GHz.

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(RE102)

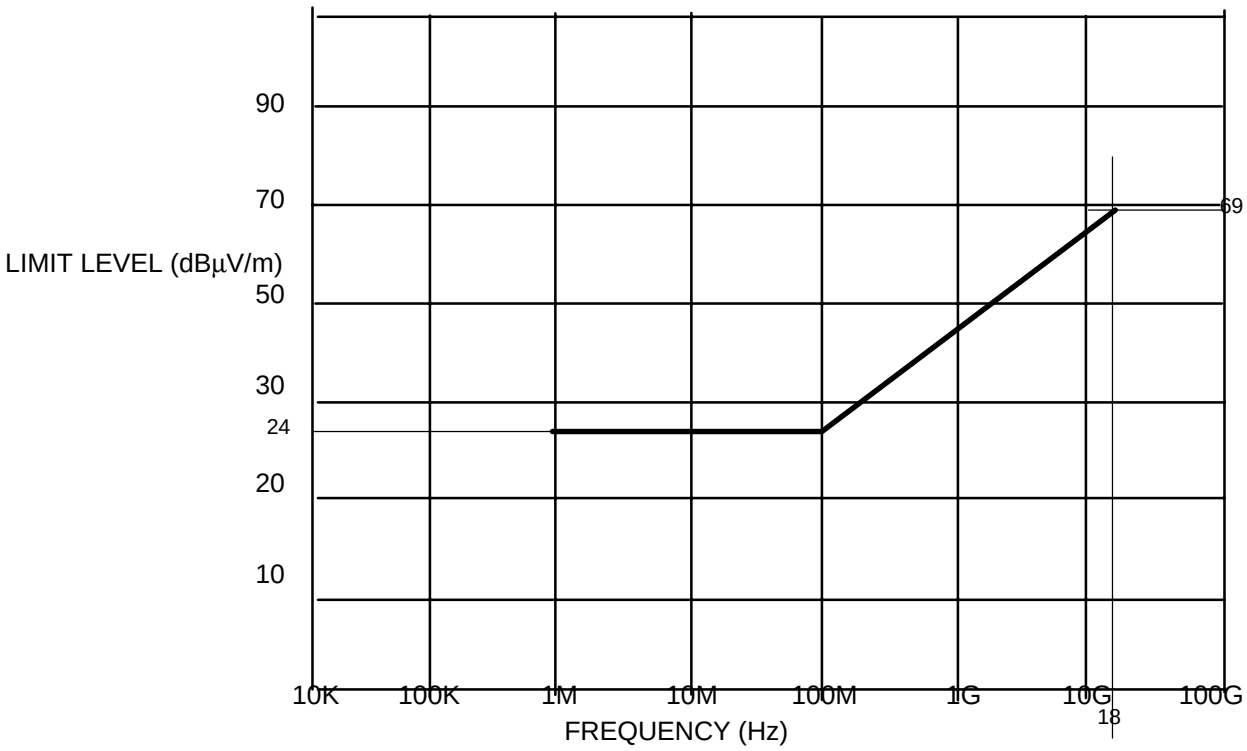


Figure B-7. Radiated emission limit, electric field.

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APPENDIX B

(RS103)

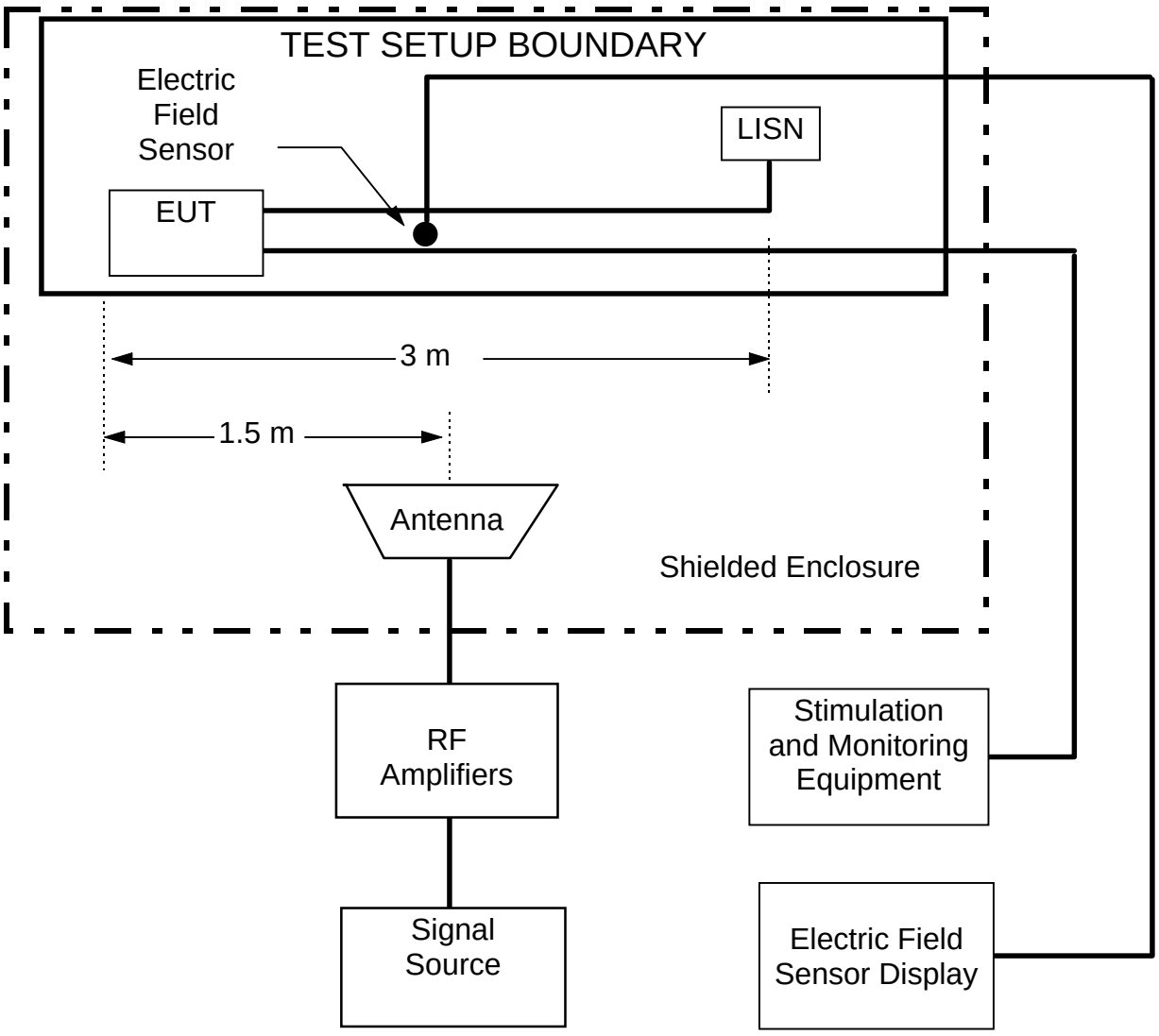


FIGURE B-8. Test setup, radiated susceptibility, electric field, 10 KHz to 40 GHz.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-49185B	**2. DOCUMENT DATE (YYMMDD)** 970519

3. DOCUMENT TITLE RADIO SET, AN/GRC-103(V)3

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7. DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME ROGER LOUISAIRE	b. TELEPHONE *Include Area Code)* (1) Commercial: (732) 532-2735 (2) AUTOVON: 992-2735
c. ADDRESS *(Include Zip Code)* US ARMY COMMUNICATIONS-ELECTRONICS COMMAND, ATTN: AMSEL-LC-LEO-E-EP FORT MONMOUTH, NJ 07703-5023	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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